

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7988
Nombre de lignes df clean	7988

Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5591
Taille dataset test	2396
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

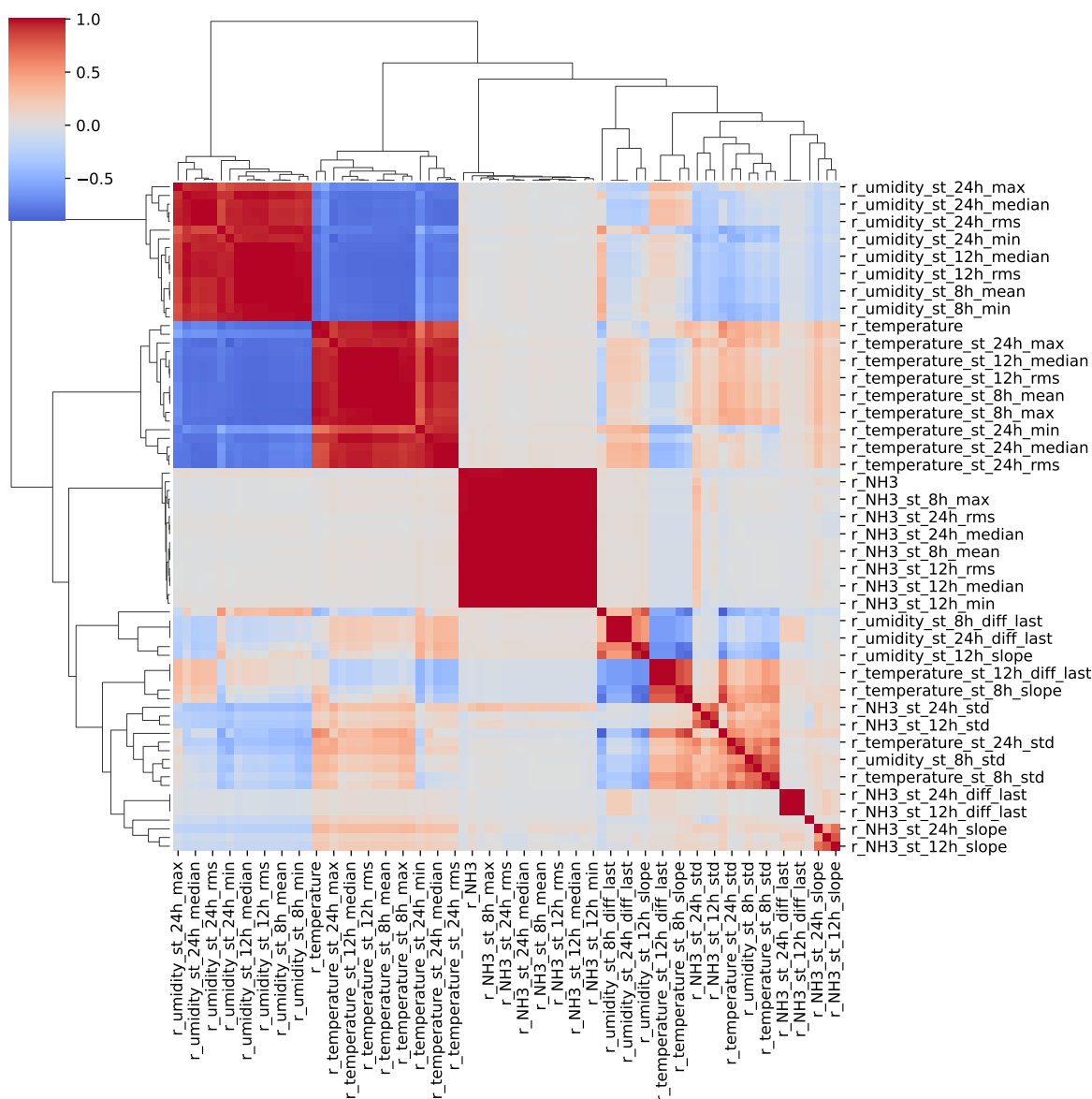
The following section has been done with a threshold choosen of : 0.8

cluster 1: r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_24h_max, r_NH3_st_12h_min, r_NH3, r_NH3_st_24h_min, r_NH3_st_8h_median, r_NH3_st_12h_median, r_NH3_st_12h_max, r_NH3_st_8h_rms, r_NH3_st_24h_mean, r_NH3_st_12h_rms, r_NH3_st_24h_rms, r_NH3_st_12h_mean, r_NH3_st_8h_mean, r_NH3_st_24h_median ; **cluster 2:** r_temperature_st_24h_median, r_temperature_st_8h_min, r_umidity_st_12h_median, r_umidity_st_24h_mean, r_temperature_st_24h_mean, r_temperature, r_umidity_st_24h_rms, r_THI, r_temperature_st_24h_min, r_temperature_st_12h_min, r_umidity_st_24h_min, r_temperature_st_12h_mean, r_umidity_st_8h_max, r_temperature_st_24h_rms, r_temperature_st_24h_max, r_umidity_st_8h_mean, r_temperature_st_8h_rms, r_umidity_st_12h_rms, r_temperature_st_8h_mean, r_umidity_st_8h_rms, r_umidity_st_8h_min, r_umidity_st_12h_min, r_umidity, r_umidity_st_24h_max, r_umidity_st_8h_median, r_temperature_st_12h_median, r_umidity_st_12h_max, r_temperature_st_8h_median, r_temperature_st_12h_rms, r_temperature_st_8h_max, r_umidity_st_12h_mean, r_temperature_st_12h_max, r_umidity_st_24h_median ; **cluster 3:** r_CO2 ; **cluster 4:** r_NH3_st_12h_std, r_NH3_st_8h_std ; **cluster 5:** r_NH3_st_24h_diff_last, r_NH3_st_8h_diff_last, r_NH3_st_12h_diff_last ; **cluster 6:** r_NH3_st_8h_slope ; **cluster 7:** r_NH3_st_12h_slope ; **cluster 8:** r_NH3_st_24h_std ; **cluster 9:** r_NH3_st_24h_slope ; **cluster 10:** r_temperature_st_8h_std, r_temperature_st_12h_std ; **cluster 11:** r_umidity_st_12h_slope, r_temperature_st_12h_diff_last, r_temperature_st_8h_diff_last, r_umidity_st_24h_slope, r_temperature_st_8h_slope, r_umidity_st_8h_slope, r_temperature_st_24h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_slope ; **cluster 12:** r_umidity_st_24h_std, r_temperature_st_24h_std ; **cluster 13:** r_umidity_st_8h_std, r_umidity_st_12h_std ; **cluster 14:** r_umidity_st_8h_diff_last, r_umidity_st_24h_diff_last, r_umidity_st_12h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3_st_8h_min, r_temperature_st_24h_median, r_CO2, r_NH3_st_12h_std, r_NH3_st_24h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_slope, r_NH3_st_24h_std, r_NH3_st_24h_slope, r_temperature_st_8h_std, r_umidity_st_12h_slope, r_umidity_st_24h_std, r_umidity_st_8h_std, r_umidity_st_8h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3_st_8h_min, r_temperature_st_24h_median, r_CO2, r_NH3_st_12h_std, r_NH3_st_24h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_slope, r_NH3_st_24h_std,

r_NH3_st_24h_slope, r_temperature_st_8h_std, r_umidity_st_12h_slope, r_umidity_st_24h_std, r_umidity_st_8h_std, r_umidity_st_8h_diff_last

Len(Features_X): 14

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

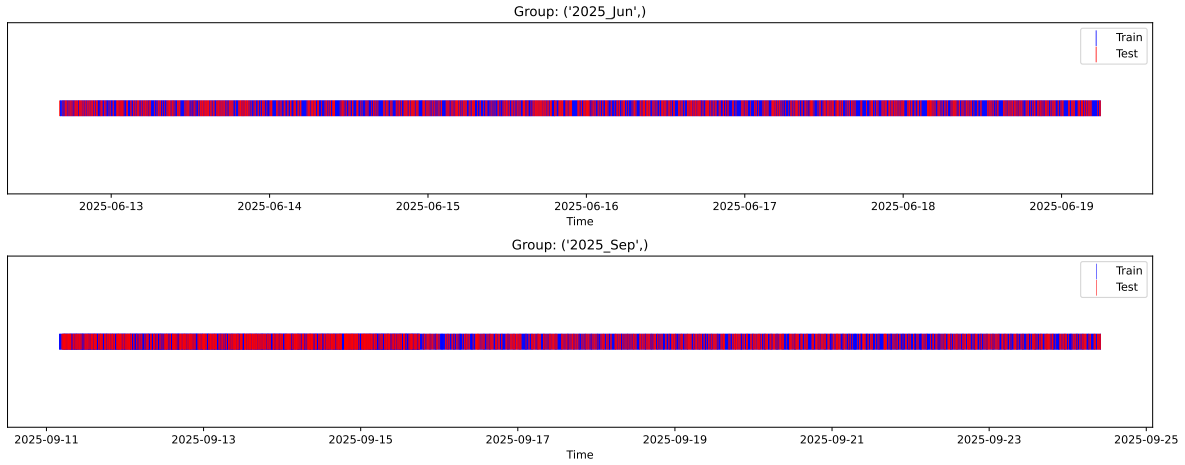
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

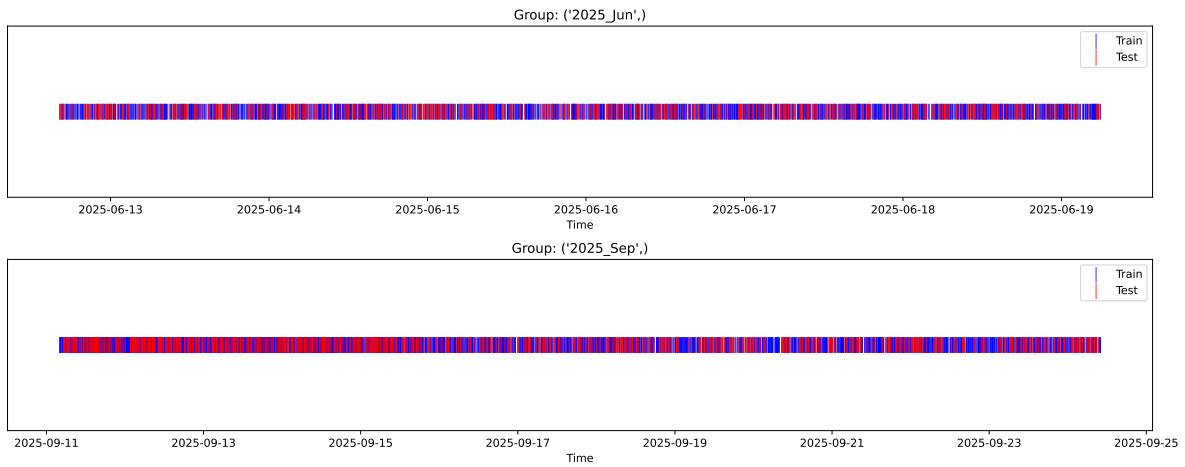
External Split: Train: 5591 samples Test: 2397 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



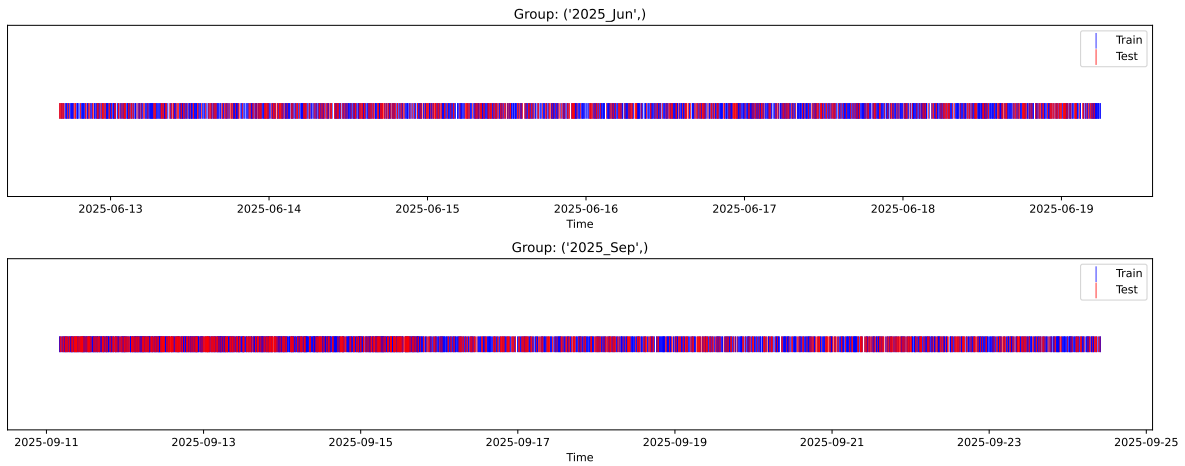
Fold 4/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.350, std=0.793, min=0.959, max=8.682
 y_{test} : mean=2.359, std=0.815, min=0.973, max=13.845

Fold 0 — y_{test} : mean=2.36, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 1 — y_{test} : mean=2.35, std=0.78, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.37, std=0.83, y_train: mean=2.34, std=0.77

Fold 4 — y_test: mean=2.37, std=0.79, y_train: mean=2.34, std=0.79

train scores CV: [0.5092921 0.49410993 0.4962565 0.49662042 0.50477527]

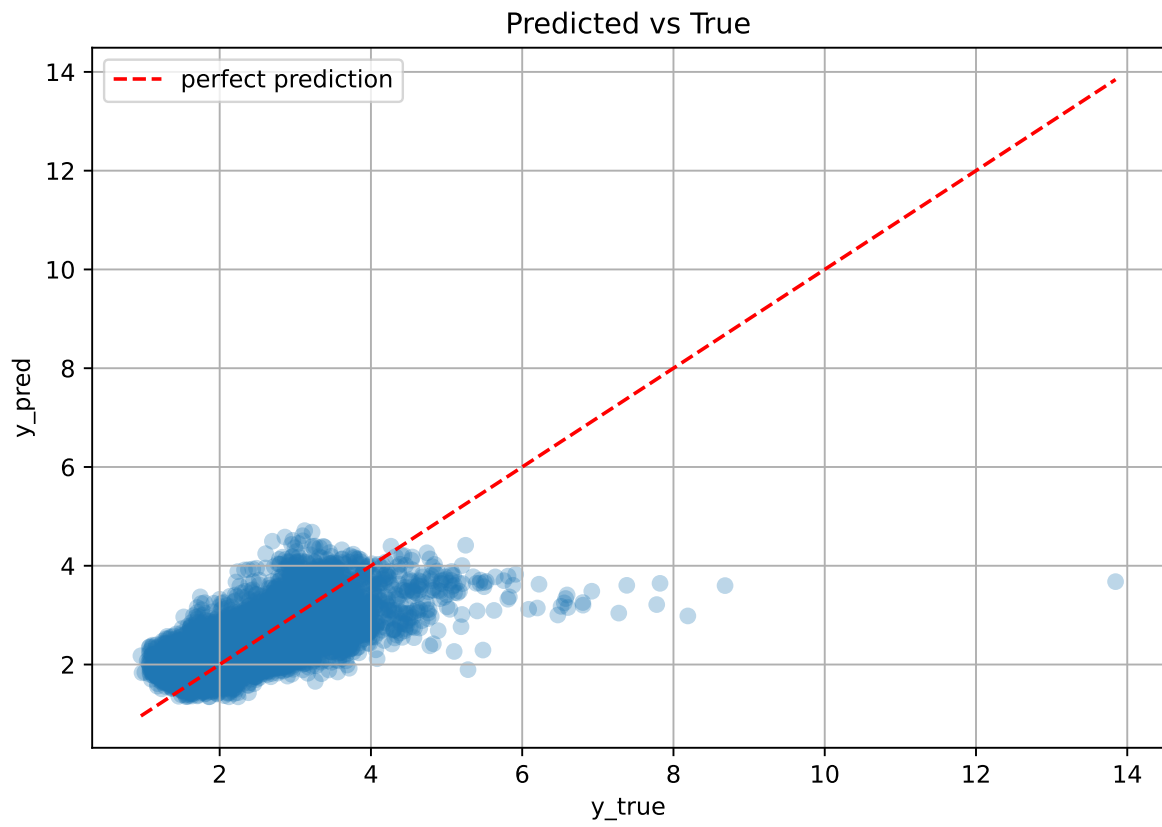
test scores CV: [0.48169226 0.51627232 0.51270811 0.50957814 0.4916542]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5002	0.5024	0.0134	0.4802	0.4806	-0.0218	0.9566	0.9957

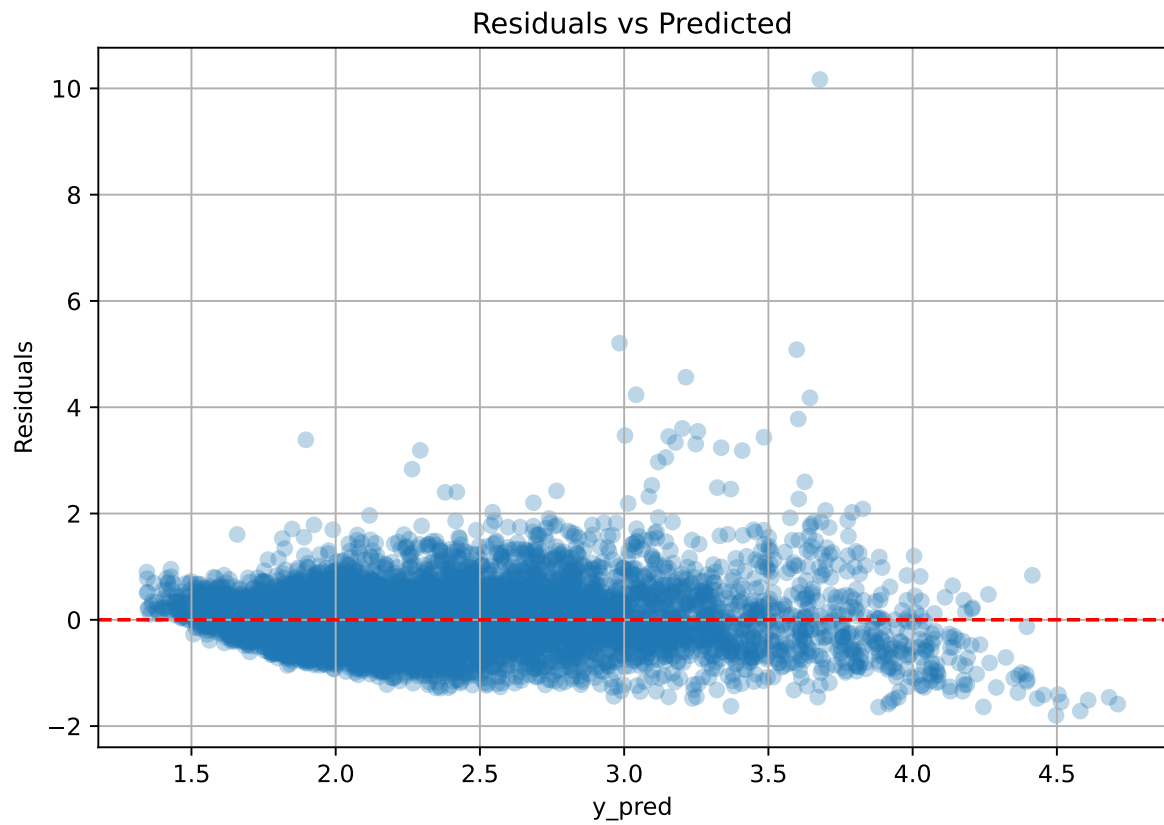
Coefficients:

coef_r_CO2 = -0.0255, coef_r_NH3_st_12h_slope = 0.0191, coef_r_NH3_st_12h_std = 0.0648, coef_r_NH3_st_24h_diff_last = 0.0254, coef_r_NH3_st_24h_slope = 0.0257, coef_r_NH3_st_24h_std = 0.0699, coef_r_NH3_st_8h_min = -0.0748, coef_r_NH3_st_8h_slope = 0.0223, coef_r_temperature_st_24h_median = 0.2477, coef_r_temperature_st_8h_std = 0.1406, coef_r_umidity_st_12h_slope = -0.3222, coef_r_umidity_st_24h_std = -0.0193, coef_r_umidity_st_8h_diff_last = -0.1093, coef_r_umidity_st_8h_std = 0.0812, intercept = 2.3500,

3.4.2 Scatter



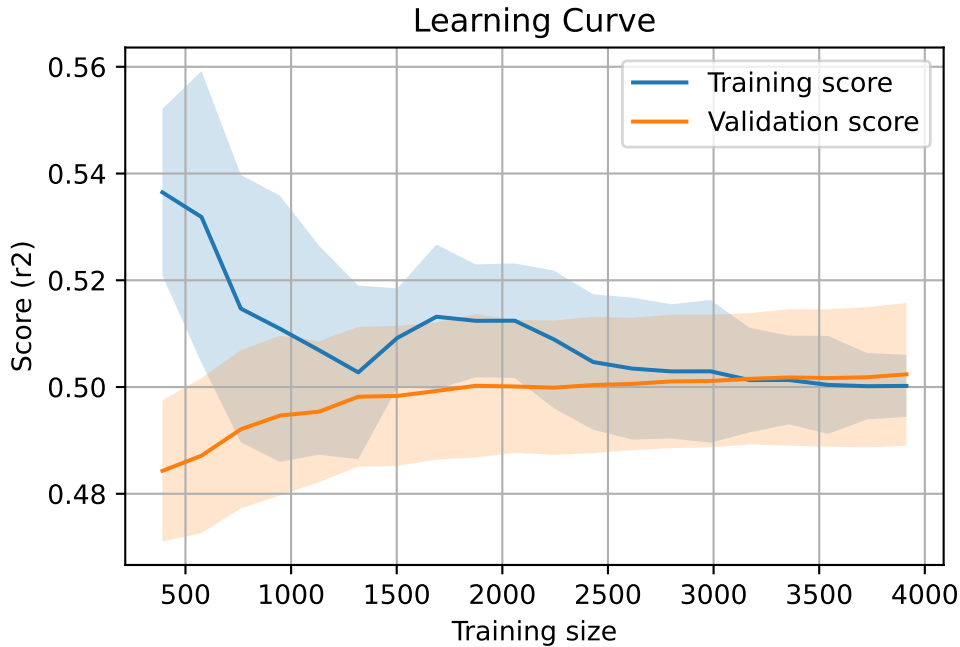
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396

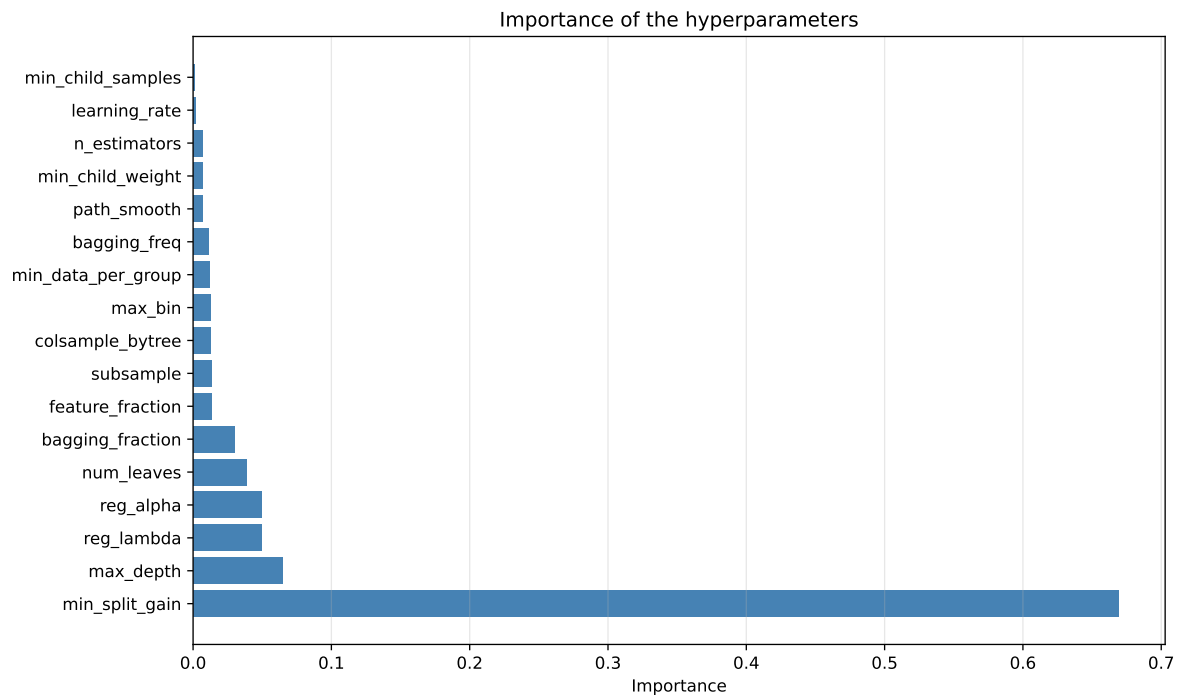
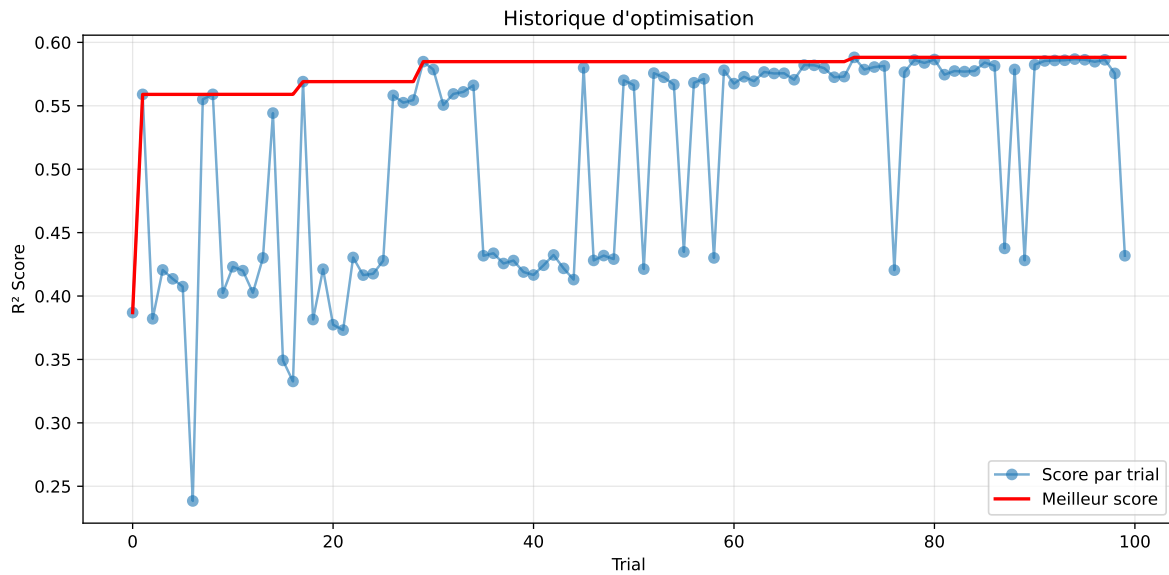


3.5 Model : LGBM

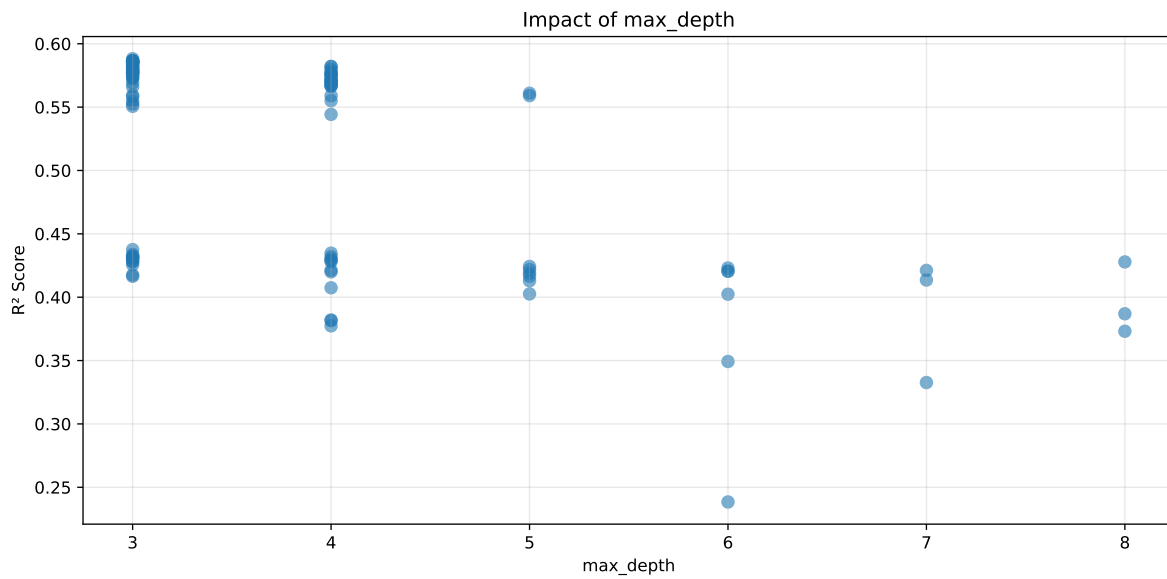
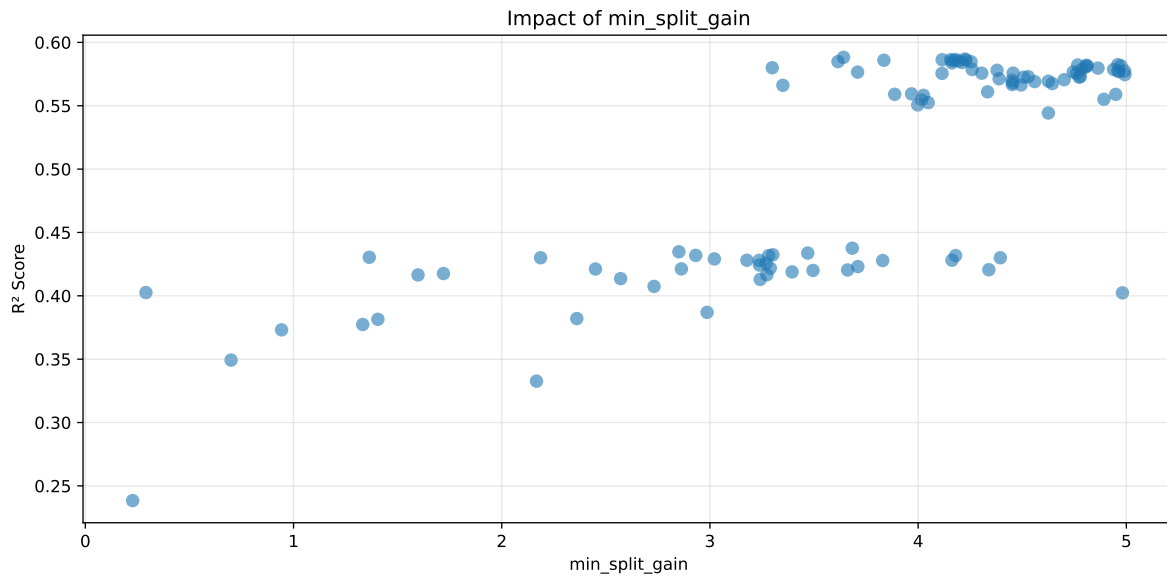
3.5.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====
feature importance r_CO2 144 r_umidity_st_12h_slope 123 r_temperature_st_24h_me-
dian 118 r_NH3_st_24h_std 79 r_temperature_st_8h_std 73 r_NH3_st_8h_min
70 r_umidity_st_8h_diff_last 60 r_NH3_st_12h_std 57 r_umidity_st_24h_std
37 r_NH3_st_24h_diff_last 35 r_NH3_st_24h_slope 30 r_NH3_st_8h_slope 17
r_NH3_st_12h_slope 17 r_umidity_st_8h_std 17

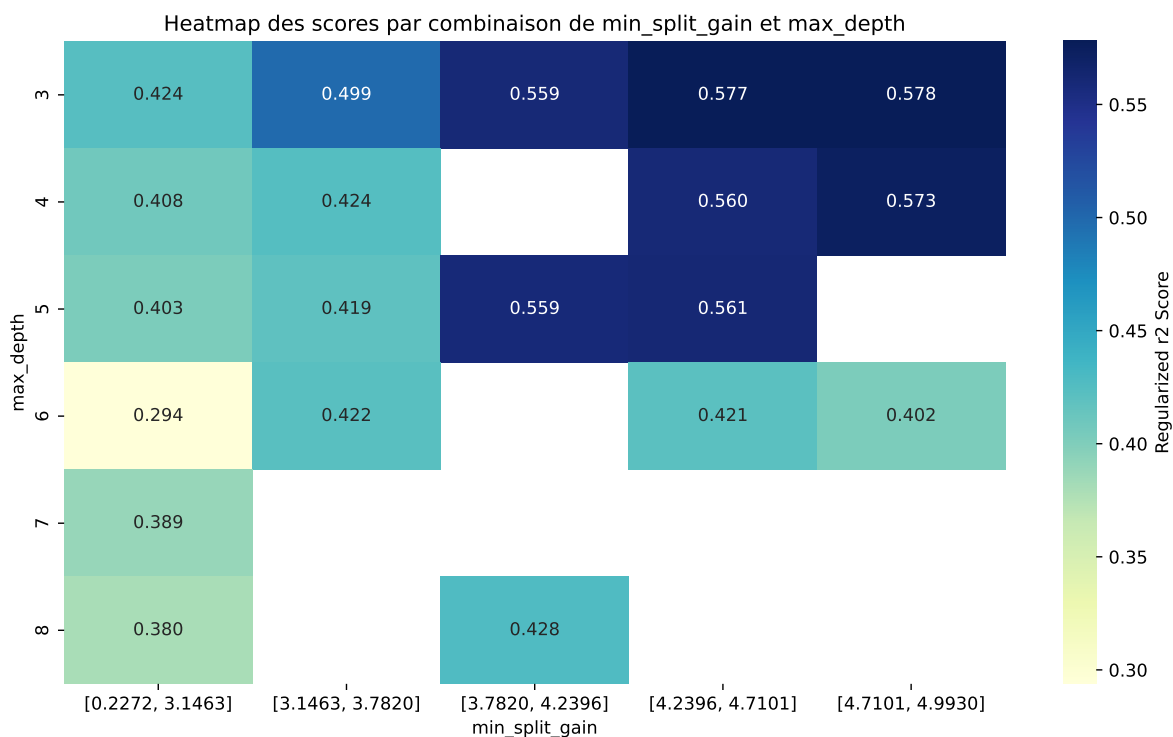


Paramètres avec >5% d'importance : ['min_split_gain', 'max_depth']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6168	0.5882	0.5738	0.0101	0.5882	0.5881	-0.0001	0.9998	1.0486
2	0.6157	0.5868	0.5717	0.0103	0.5868	0.5855	-0.0014	0.9977	1.0491
3	0.6143	0.5865	0.5704	0.0107	0.5865	0.5848	-0.0017	0.9972	1.0474
4	0.6132	0.5863	0.5701	0.0097	0.5863	0.5817	-0.0045	0.9922	1.0459
5	0.6129	0.5863	0.5703	0.0117	0.5863	0.585	-0.0013	0.9978	1.0454
6	0.6133	0.5861	0.569	0.0106	0.5861	0.586	-0.0001	0.9999	1.0464
7	0.6146	0.5859	0.5737	0.0099	0.5859	0.5872	0.0013	1.0022	1.0489
8	0.612	0.5857	0.5688	0.0112	0.5857	0.5832	-0.0025	0.9957	1.0449
9	0.61	0.5854	0.5683	0.0106	0.5854	0.5807	-0.0047	0.9919	1.0421
10	0.614	0.5848	0.5685	0.0104	0.5848	0.5829	-0.0019	0.9967	1.0499
---	---	---	---	---	---	---	---	---	---
100	0.743	0.6589	0.6553	0.0086	0.2384	0.6802	0.0213	1.0323	1.1276

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.03182864505600817, 'reg__num_leaves': 60, 'reg__subsample': 0.5686853751293445, 'reg__colsample_bytree': 0.8984935000075815, 'reg__min_child_samples': 31, 'reg__reg_alpha': 1.1814343026834173, 'reg__reg_lambda': 41.92767289919442, 'reg__min_split_gain': 3.641808214356309, 'reg__path_smooth': 1.776969919178301, 'reg__min_child_weight':

0.05491724841396077, 'reg__feature_fraction': 0.821264305207929, 'reg__bagging_fraction': 0.7019849029111352, 'reg__bagging_freq': 1, 'reg__max_bin': 96, 'reg__min_data_per_group': 194}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.046458133592947856, 'reg__num_leaves': 63, 'reg__subsample': 0.618168535734279, 'reg__colsample_bytree': 0.7448895819952778, 'reg__min_child_samples': 41, 'reg__reg_alpha': 0.0016463261718260291, 'reg__reg_lambda': 24.596542470262932, 'reg__min_split_gain': 4.223942566133249, 'reg__path_smooth': 0.9467206959890756, 'reg__min_child_weight': 0.025426652943588826, 'reg__feature_fraction': 0.6875060266501489, 'reg__bagging_fraction': 0.6659608335591722, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 171}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.04839993104802782, 'reg__num_leaves': 52, 'reg__subsample': 0.6274593879790171, 'reg__colsample_bytree': 0.7580068402575761, 'reg__min_child_samples': 39, 'reg__reg_alpha': 0.38259818038195004, 'reg__reg_lambda': 49.95651607263662, 'reg__min_split_gain': 4.179619558231602, 'reg__path_smooth': 1.1811914160927057, 'reg__min_child_weight': 0.030149373493218556, 'reg__feature_fraction': 0.7220412061327881, 'reg__bagging_fraction': 0.8516992022570739, 'reg__bagging_freq': 1, 'reg__max_bin': 87, 'reg__min_data_per_group': 158}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.08263299600771615, 'reg__num_leaves': 55, 'reg__subsample': 0.6192924564929423, 'reg__colsample_bytree': 0.7228201486216086, 'reg__min_child_samples': 44, 'reg__reg_alpha': 0.02602899354336674, 'reg__reg_lambda': 45.722450570592414, 'reg__min_split_gain': 4.1153661442027065, 'reg__path_smooth': 0.7137716036258943, 'reg__min_child_weight': 0.027553221788827836, 'reg__feature_fraction': 0.6813614045334295, 'reg__bagging_fraction': 0.659573979852275, 'reg__bagging_freq': 1, 'reg__max_bin': 70, 'reg__min_data_per_group': 169}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.06887086927509217, 'reg__num_leaves': 55, 'reg__subsample': 0.5315298751923893, 'reg__colsample_bytree': 0.7148878095729331, 'reg__min_child_samples': 44, 'reg__reg_alpha': 0.002378944194997734, 'reg__reg_lambda': 40.176411575039815, 'reg__min_split_gain': 4.156035161975848, 'reg__path_smooth': 0.9169699696579249, 'reg__min_child_weight': 0.03019217908904485, 'reg__feature_fraction': 0.6822555368881591, 'reg__bagging_fraction': 0.6666942982939279, 'reg__bagging_freq': 1, 'reg__max_bin': 74, 'reg__min_data_per_group': 169}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.04820489320729702, 'reg__num_leaves': 51, 'reg__subsample': 0.6286838320747152, 'reg__colsample_bytree': 0.8966923276701952, 'reg__min_child_samples': 38, 'reg__reg_alpha': 1.0025299358086182, 'reg__reg_lambda': 41.69814719023002, 'reg__min_split_gain': 4.229862836914416, 'reg__path_smooth': 0.7079567260324763, 'reg__min_child_weight':

0.01957763825249181, 'reg__feature_fraction': 0.7135217346951427, 'reg__bagging_fraction': 0.7575648464620525, 'reg__bagging_freq': 1, 'reg__max_bin': 87, 'reg__min_data_per_group': 195}

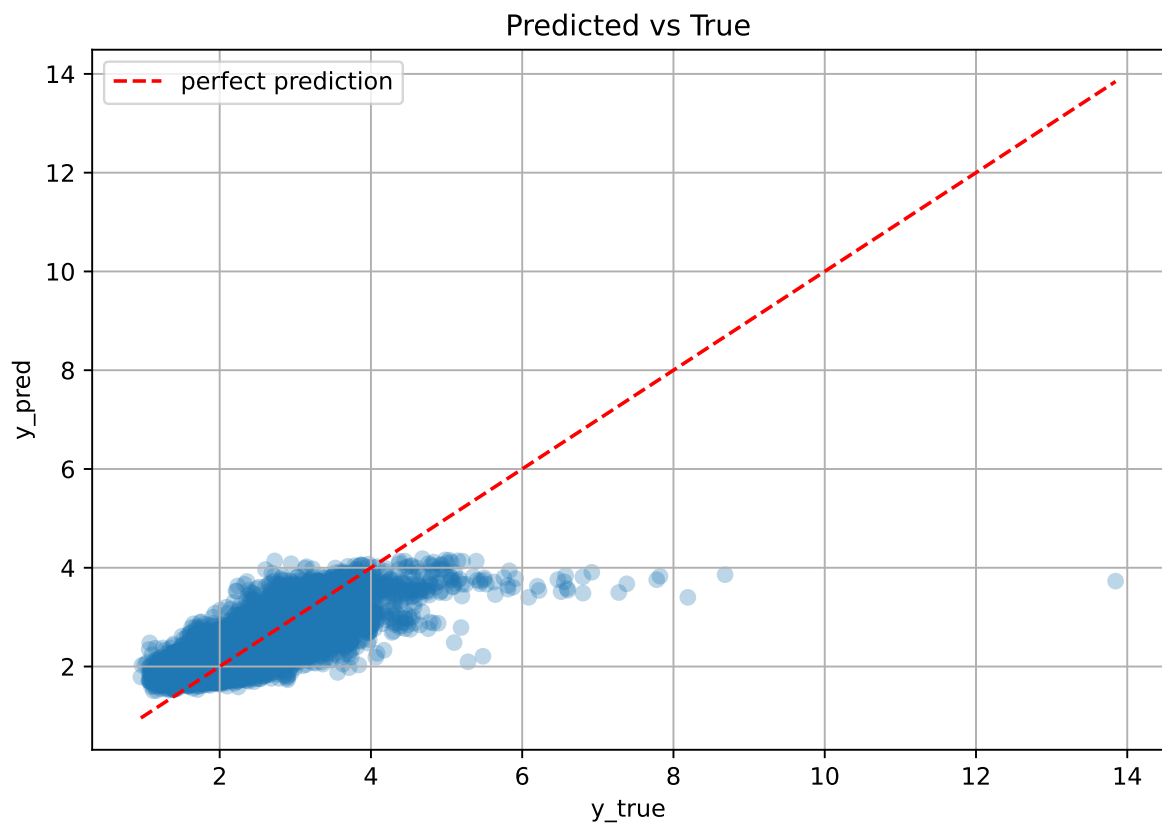
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.08816528053776251, 'reg__num_leaves': 54, 'reg__subsample': 0.6151856808745264, 'reg__colsample_bytree': 0.7528527258092436, 'reg__min_child_samples': 42, 'reg__reg_alpha': 0.44142429040024406, 'reg__reg_lambda': 49.92191821852694, 'reg__min_split_gain': 3.8355191656618723, 'reg__path_smooth': 0.7871225970918538, 'reg__min_child_weight': 0.030961797315285568, 'reg__feature_fraction': 0.8436326358397537, 'reg__bagging_fraction': 0.6687201729477866, 'reg__bagging_freq': 1, 'reg__max_bin': 69, 'reg__min_data_per_group': 143}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.08723729360594827, 'reg__num_leaves': 63, 'reg__subsample': 0.6189431349959785, 'reg__colsample_bytree': 0.7447908433890975, 'reg__min_child_samples': 42, 'reg__reg_alpha': 0.5077422202773267, 'reg__reg_lambda': 41.946751840000125, 'reg__min_split_gain': 4.168348001660865, 'reg__path_smooth': 0.8676074883234748, 'reg__min_child_weight': 0.03404479655223544, 'reg__feature_fraction': 0.6790272558143106, 'reg__bagging_fraction': 0.6815827520549853, 'reg__bagging_freq': 1, 'reg__max_bin': 65, 'reg__min_data_per_group': 195}

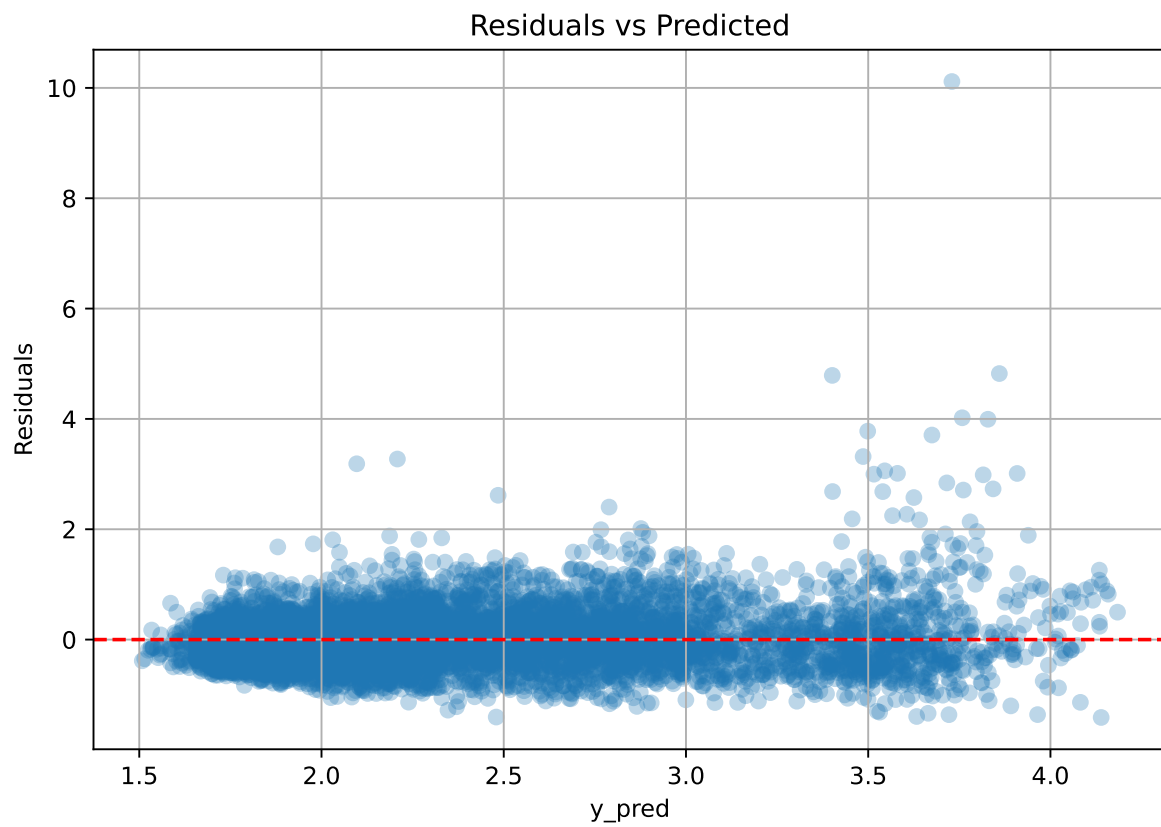
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.08159905388233231, 'reg__num_leaves': 63, 'reg__subsample': 0.6235897587766375, 'reg__colsample_bytree': 0.8965595202166438, 'reg__min_child_samples': 42, 'reg__reg_alpha': 0.4242301535043167, 'reg__reg_lambda': 45.23293763920628, 'reg__min_split_gain': 4.1933554700607, 'reg__path_smooth': 0.8974664485416963, 'reg__min_child_weight': 0.02286323806847657, 'reg__feature_fraction': 0.6824621117479985, 'reg__bagging_fraction': 0.6697489069053238, 'reg__bagging_freq': 1, 'reg__max_bin': 64, 'reg__min_data_per_group': 170}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.020729204892858736, 'reg__num_leaves': 24, 'reg__subsample': 0.5930923035474744, 'reg__colsample_bytree': 0.781959998042917, 'reg__min_child_samples': 11, 'reg__reg_alpha': 3.2017428272861572, 'reg__reg_lambda': 37.04634965822551, 'reg__min_split_gain': 3.6150438525490096, 'reg__path_smooth': 2.682678398167934, 'reg__min_child_weight': 0.45274728064658076, 'reg__feature_fraction': 0.880128266850717, 'reg__bagging_fraction': 0.777854364774818, 'reg__bagging_freq': 1, 'reg__max_bin': 144, 'reg__min_data_per_group': 196}

3.5.2 Scatter



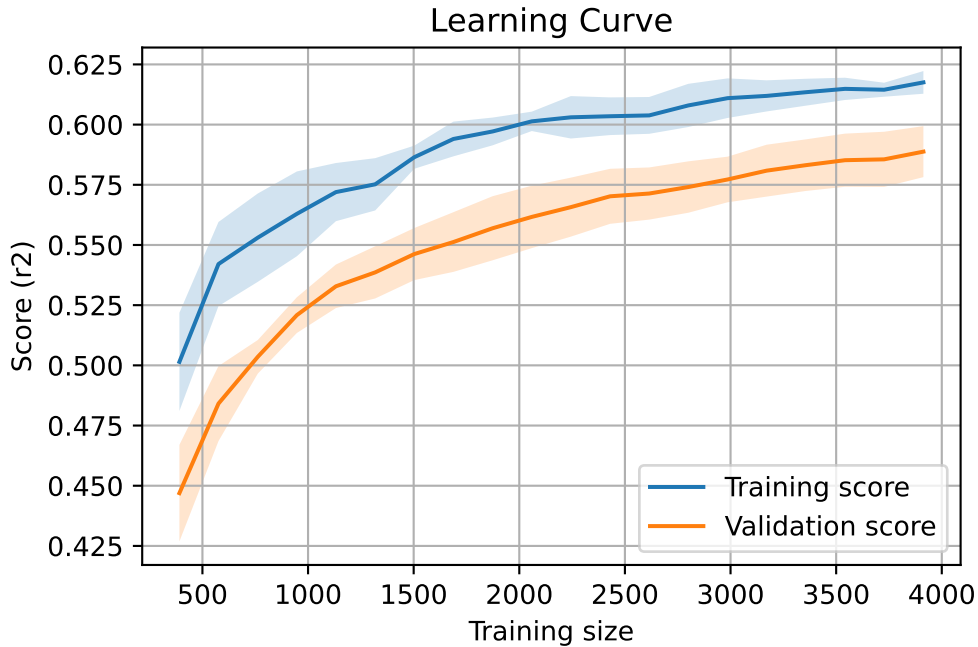
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



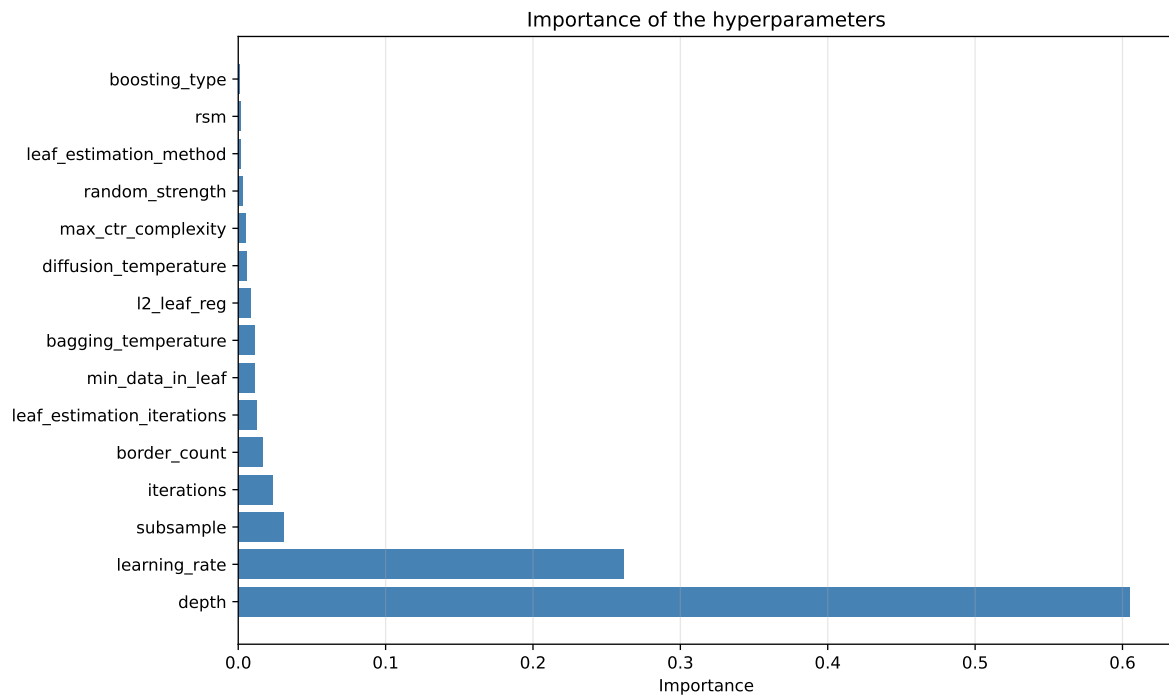
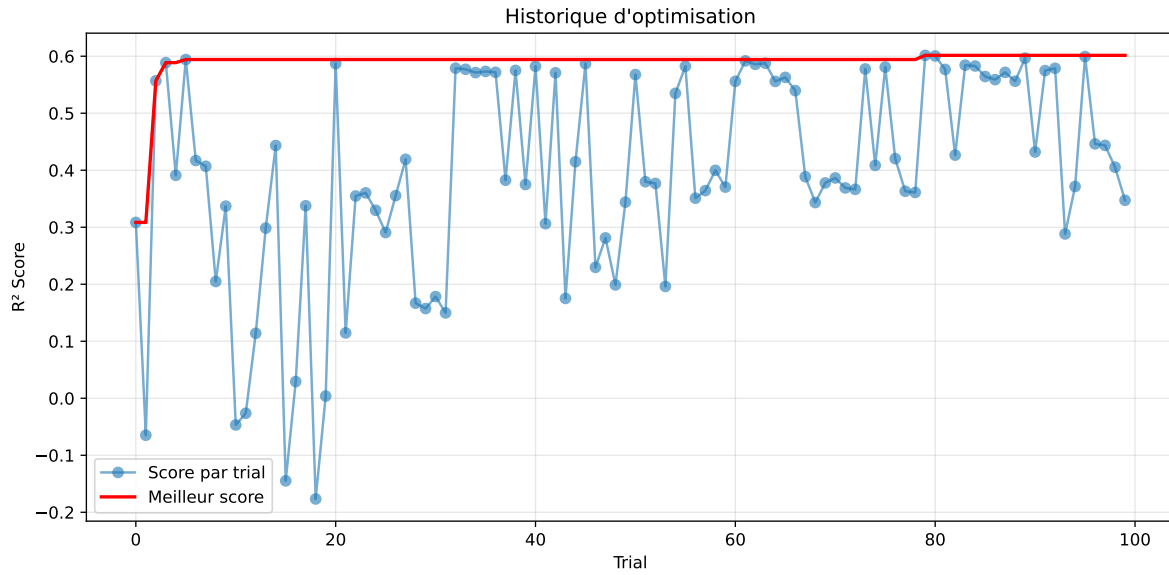
3.6 Model : CatBoost

3.6.1 Gridsearch

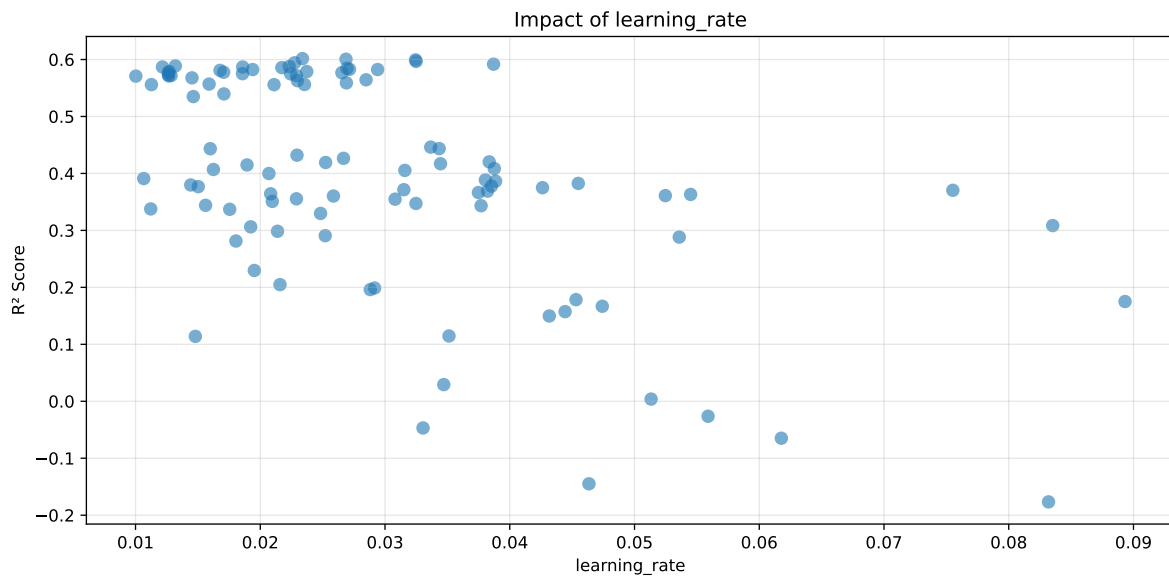
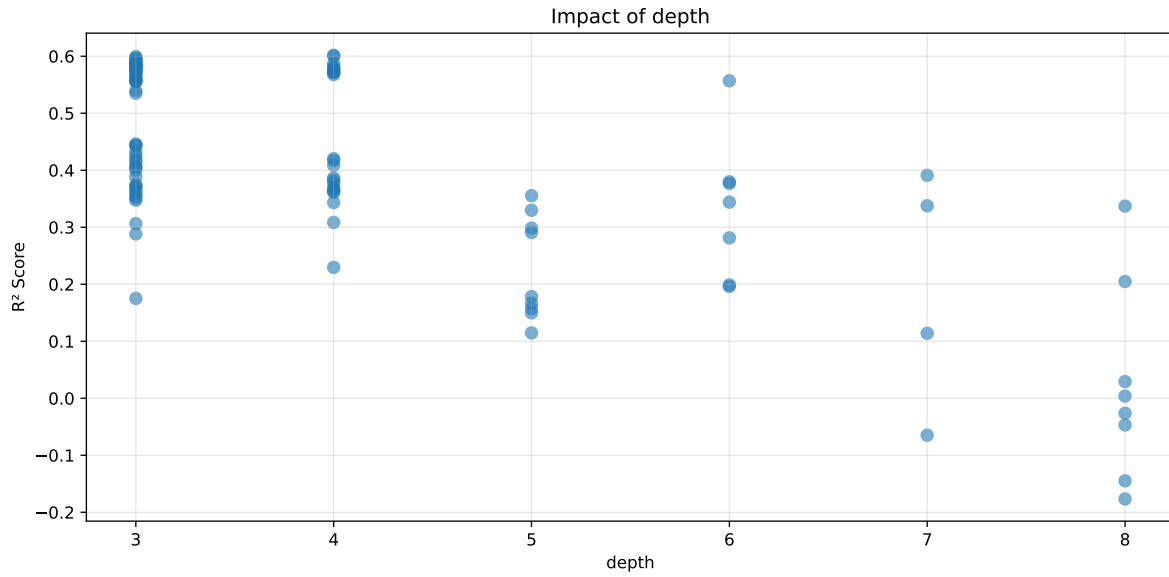
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

===== TOP Importance FEATURES =====

feature	importance
r_umidity_st_12h_slope	31.999155
r_temperature_st_24h_median	17.334770
r_temperature_st_8h_std	11.884920
r_NH3_st_8h_min	6.756767
r_CO2	6.018289
r_umidity_st_8h_diff_last	4.624322
r_NH3_st_12h_std	4.613179
r_NH3_st_24h_std	3.934105
r_NH3_st_24h_slope	2.698278
r_NH3_st_8h_slope	2.390948
r_NH3_st_12h_slope	2.328557
r_umidity_st_24h_std	2.152429
r_umidity_st_8h_std	1.846560
r_NH3_st_24h_diff_last	1.417722

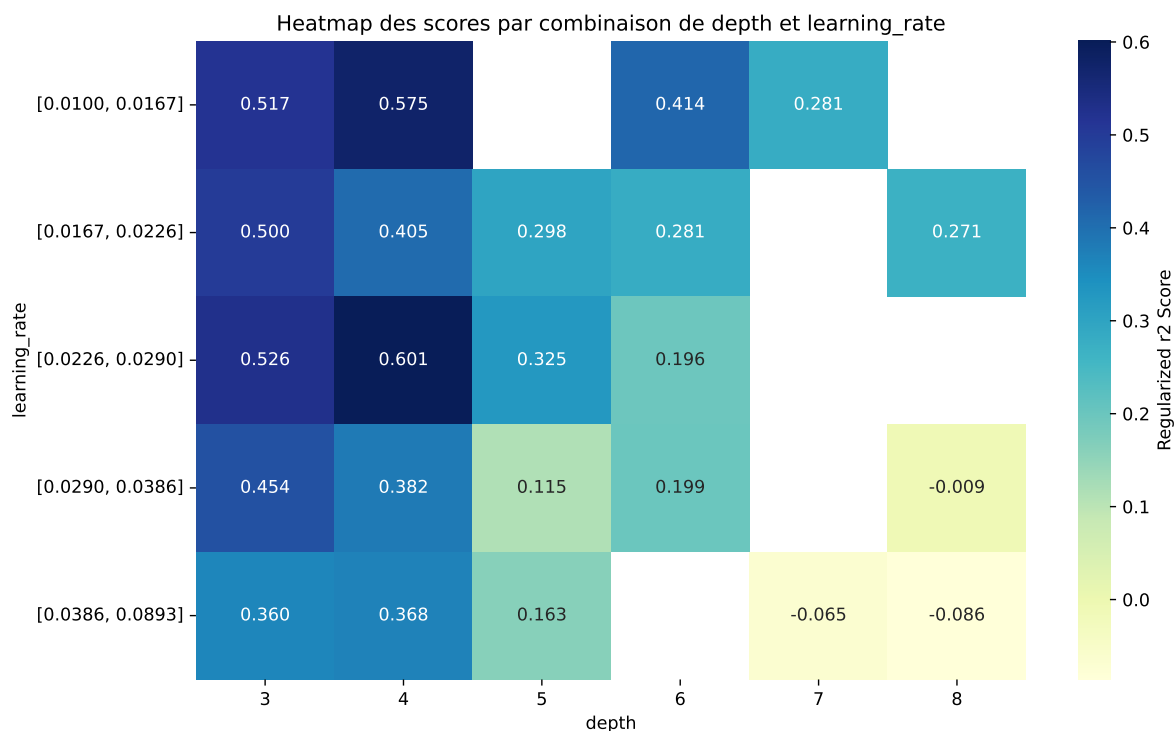


Paramètres avec >5% d'importance : ['depth', 'learning_rate']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6301	0.6015	0.5894	0.0097	0.6015	0.5953	-0.0062	0.9897	1.0475
2	0.6278	0.6005	0.5893	0.0103	0.6005	0.595	-0.0055	0.9908	1.0456
3	0.6278	0.5994	0.5854	0.0089	0.5994	0.5883	-0.0111	0.9815	1.0474
4	0.6236	0.5967	0.5833	0.0096	0.5967	0.5881	-0.0085	0.9857	1.0452
5	0.6169	0.5942	0.58	0.0108	0.5942	0.5823	-0.0119	0.9799	1.0382
6	0.6118	0.5918	0.5807	0.011	0.5918	0.5856	-0.0062	0.9896	1.0338
7	0.6104	0.5887	0.5735	0.0096	0.5887	0.5758	-0.0129	0.9781	1.0368
8	0.6098	0.5877	0.5731	0.0098	0.5877	0.5769	-0.0108	0.9816	1.0376
9	0.6115	0.587	0.5714	0.0101	0.587	0.575	-0.0119	0.9797	1.0418
10	0.6141	0.5869	0.5713	0.0087	0.5869	0.5726	-0.0143	0.9756	1.0463
---	---	---	---	---	---	---	---	---	---
100	0.9132	0.7316	0.7185	0.0089	-0.1766	0.7411	0.0096	1.0131	1.2483

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02338070448925179, 'reg__l2_leaf_reg': 11.993680415880295, 'reg__bagging_temperature': 2.197097439099903, 'reg__random_strength': 2.829690193606159, 'reg__subsample': 0.8513337490788923, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.850292087035267, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6763.084054872704}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02687053153582589, 'reg__l2_leaf_reg': 46.74008944208243, 'reg__bagging_temperature': 1.143552759331531, 'reg__random_strength': 2.90530424667196, 'reg__subsample': 0.8564777240721624, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5368368851072184, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6746.762999880351}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.032445732032085985, 'reg__l2_leaf_reg': 3.73899594707911, 'reg__bagging_temperature': 1.5399480582723615, 'reg__random_strength': 3.273033401348824, 'reg__subsample': 0.8722432098947992, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7449511335755861, 'reg__border_count': 55, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6100.440260946417}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.03249293665312909, 'reg__l2_leaf_reg': 6.291143837408974, 'reg__bagging_temperature': 2.2082966414287157, 'reg__random_strength': 3.257687433366194, 'reg__subsample': 0.8791923963578984, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7196433611780789, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2913.724668253928}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02274770035933043, 'reg__l2_leaf_reg': 45.794937819566165, 'reg__bagging_temperature': 4.394600375217728, 'reg__random_strength': 0.7843755361232807, 'reg__subsample': 0.8705520882767137, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6292253804520548, 'reg__border_count': 117, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4593.380294013734}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.03869936663535863, 'reg__l2_leaf_reg': 16.368563715902603, 'reg__bagging_temperature': 4.211653295161628, 'reg__random_strength': 1.3272485261535685, 'reg__subsample': 0.7387777211910043, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7955869712894725, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7387.375547914813}

Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.013192589787449865, 'reg__l2_leaf_reg': 49.05414602739119, 'reg__bagging_temperature': 4.974755194867776, 'reg__random_strength': 3.3811862260614327, 'reg__subsample': 0.8209044700382546,

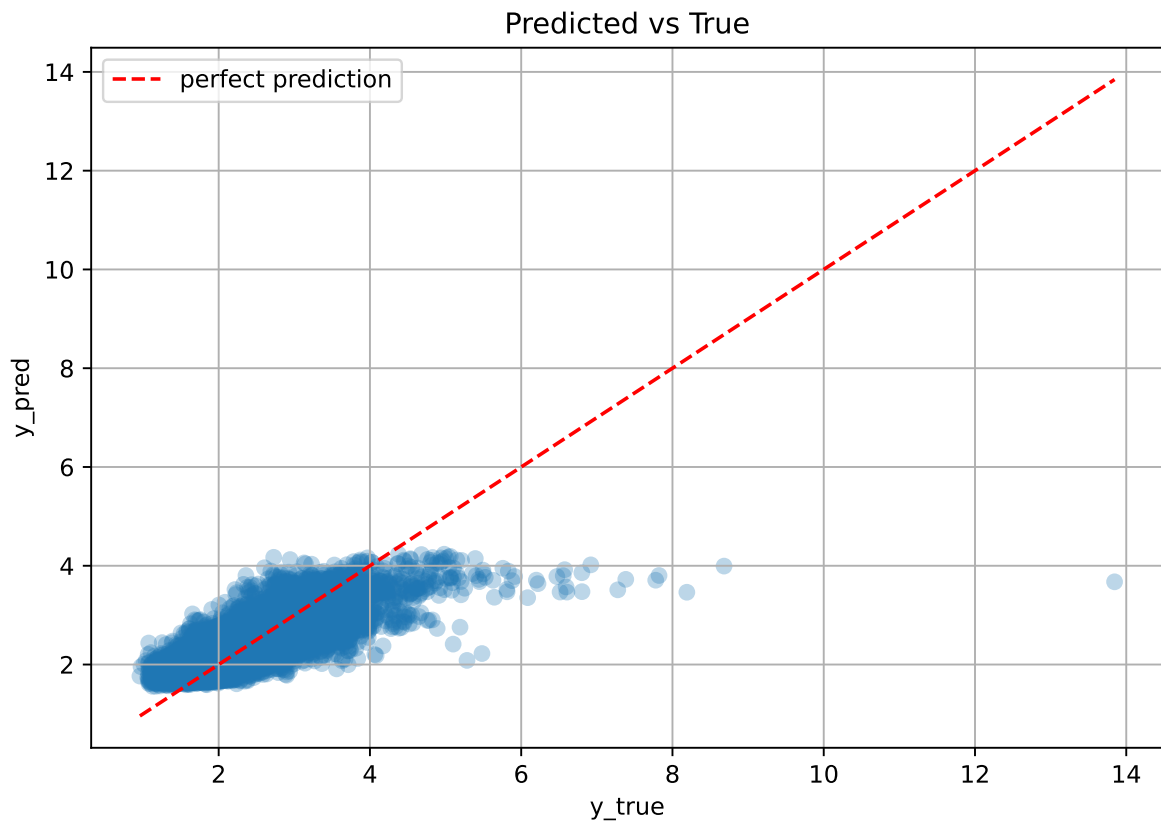
'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5260103889834293, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5887.649432308732}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.022318819604944275, 'reg__l2_leaf_reg': 15.420474724771662, 'reg__bagging_temperature': 4.220445996276731, 'reg__random_strength': 1.6955142746486225, 'reg__subsample': 0.8549040076116039, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8487573262542295, 'reg__border_count': 50, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7011.289932816171}

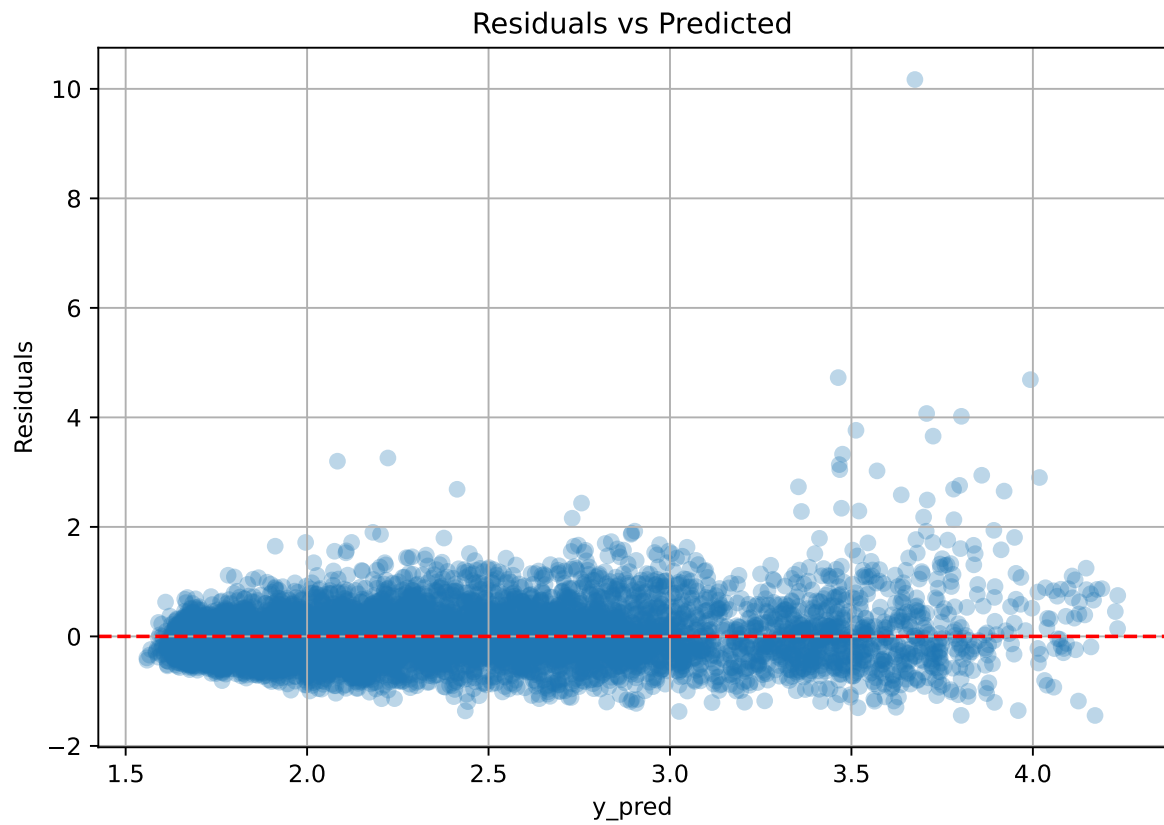
Rank 9: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.012158326944403343, 'reg__l2_leaf_reg': 20.05176466258669, 'reg__bagging_temperature': 1.73212617137247, 'reg__random_strength': 1.7083026764937737, 'reg__subsample': 0.5169787656724054, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8300678620003074, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1212.2307969005242}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01858708671957743, 'reg__l2_leaf_reg': 23.371892539361134, 'reg__bagging_temperature': 3.5791184049963443, 'reg__random_strength': 0.5988894688592074, 'reg__subsample': 0.7851372119120616, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.833931174907056, 'reg__border_count': 76, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6199.550036196055}

3.6.2 Scatter



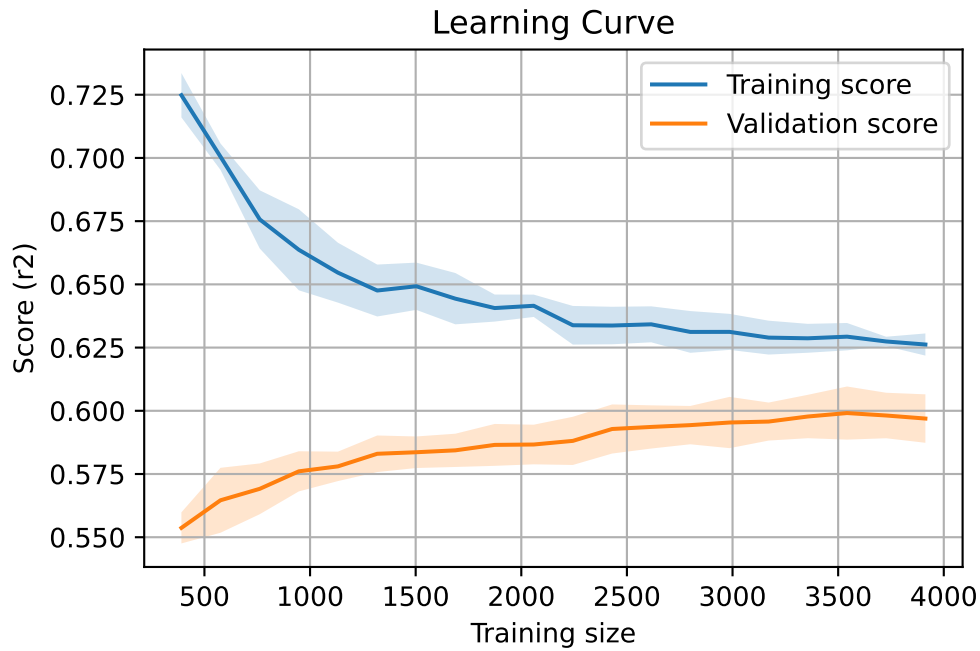
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.5	0.502	0.013	0.48	0.481	-0.022	0.957	0.996	nan
LGBM	0.617	0.588	0.01	0.574	0.588	-0.0	1.0	1.049	0.5882
CatBoost	0.63	0.602	0.01	0.589	0.595	-0.006	0.99	1.048	0.6015

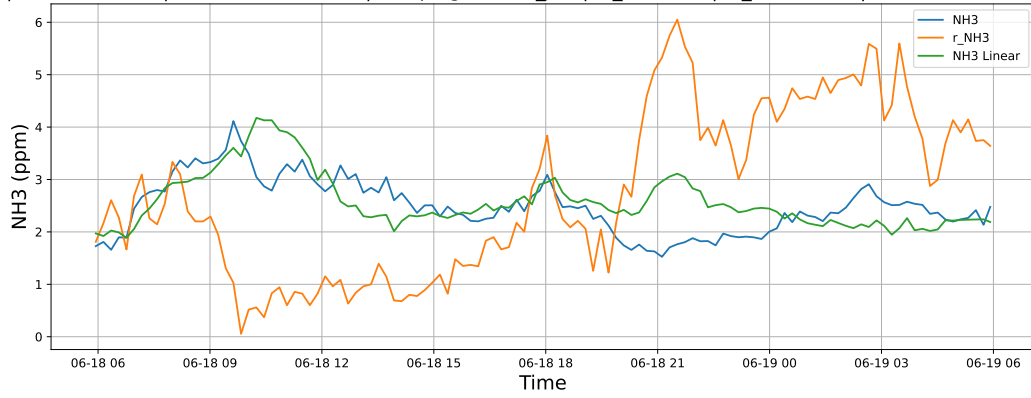
4 Plot

4.1 Standalone plots

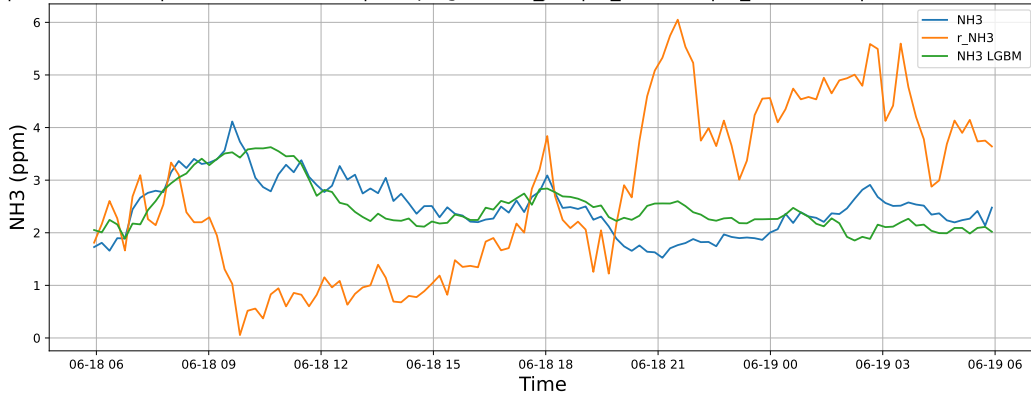
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

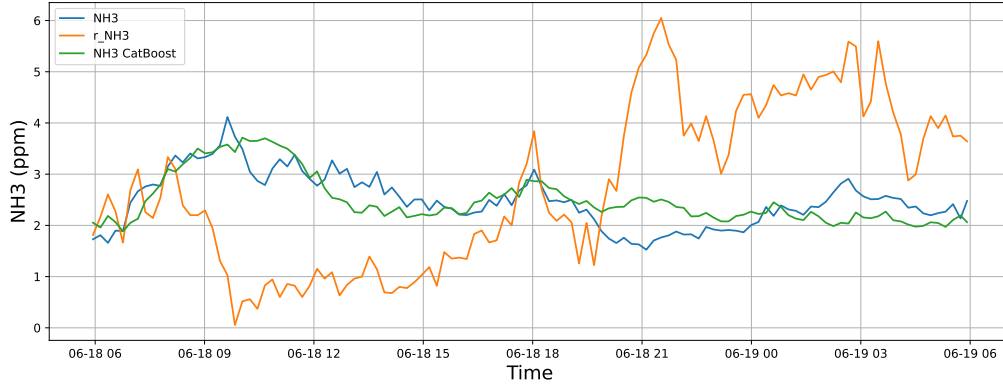
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

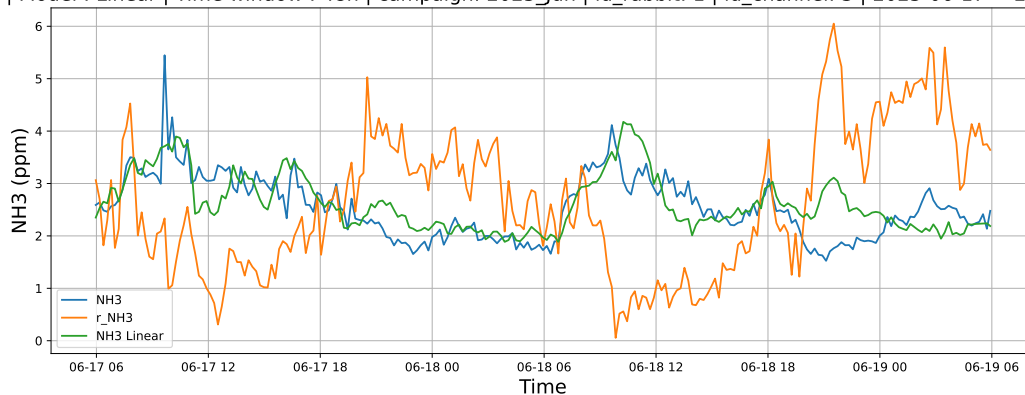


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

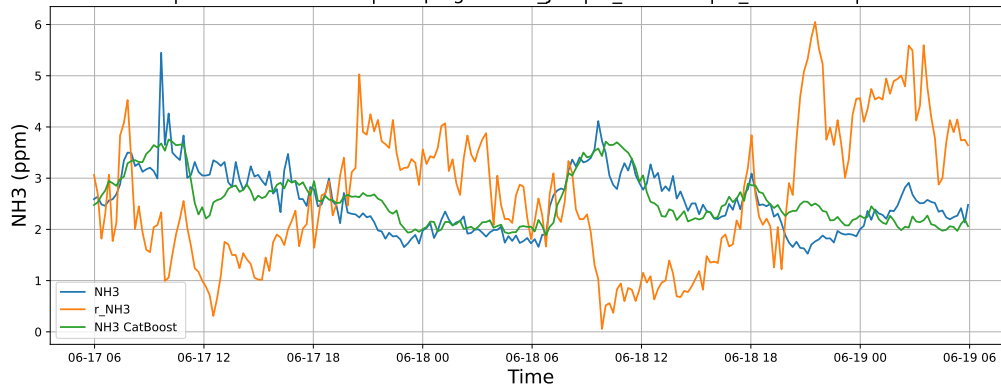
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

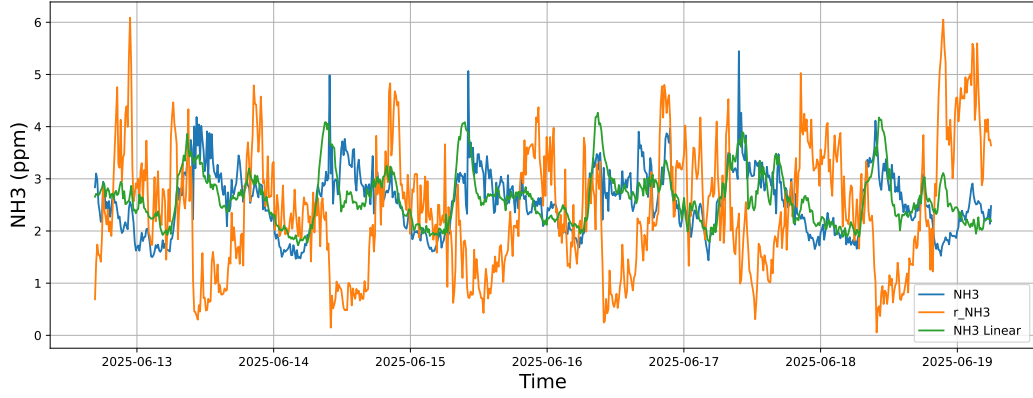


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

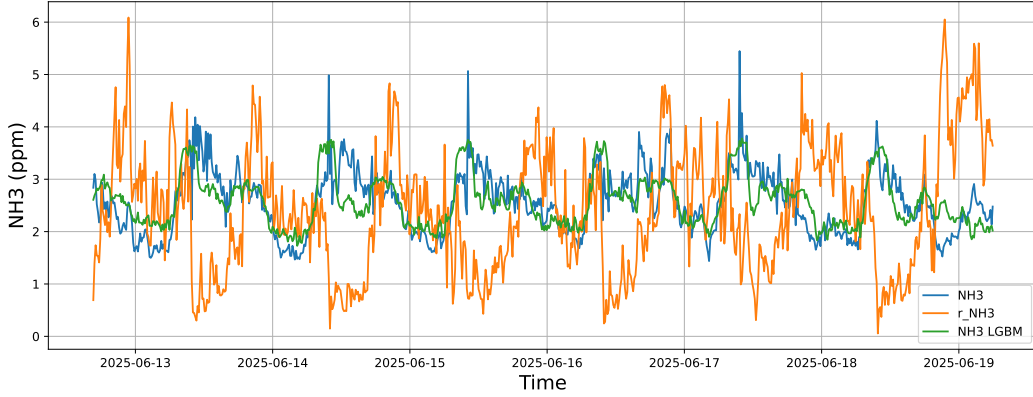


4.1.1.3 Time window : ALL

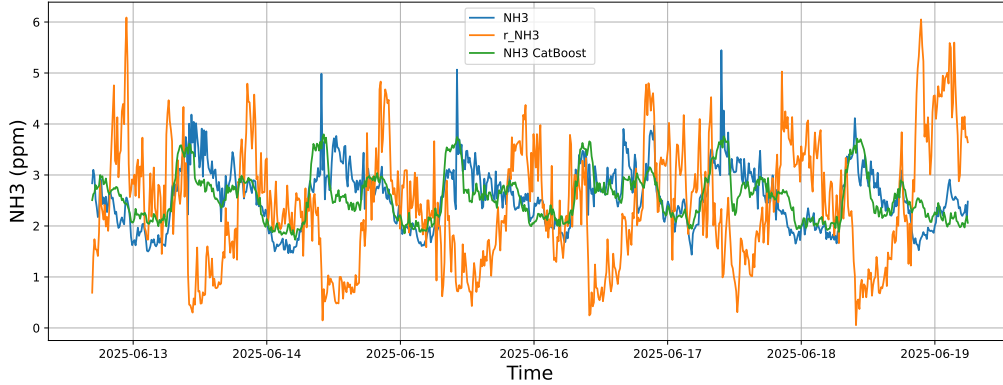
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



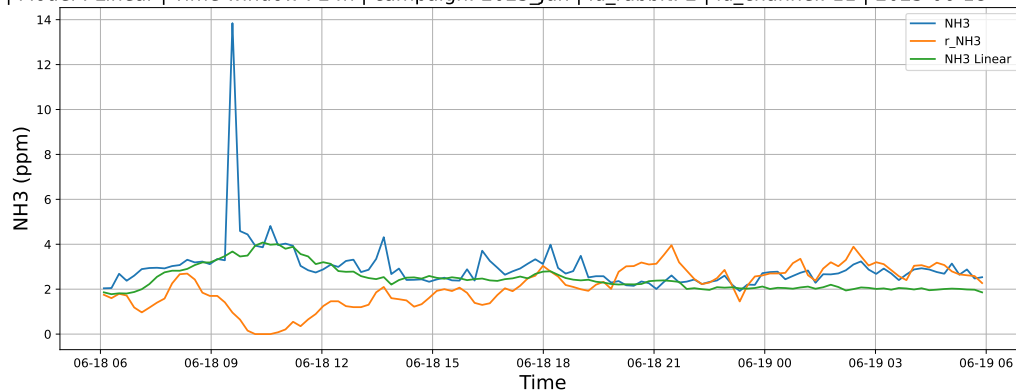
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



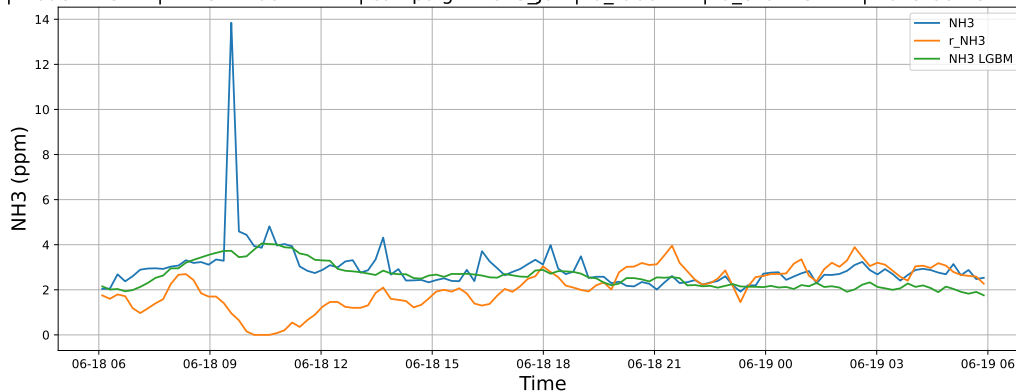
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

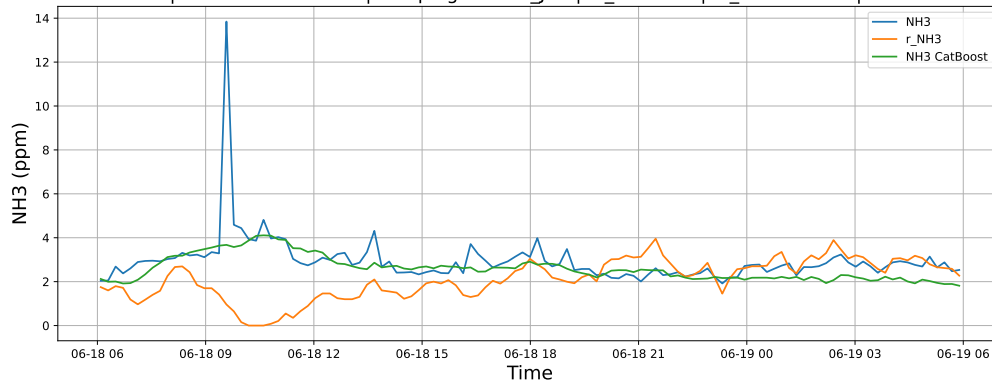
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

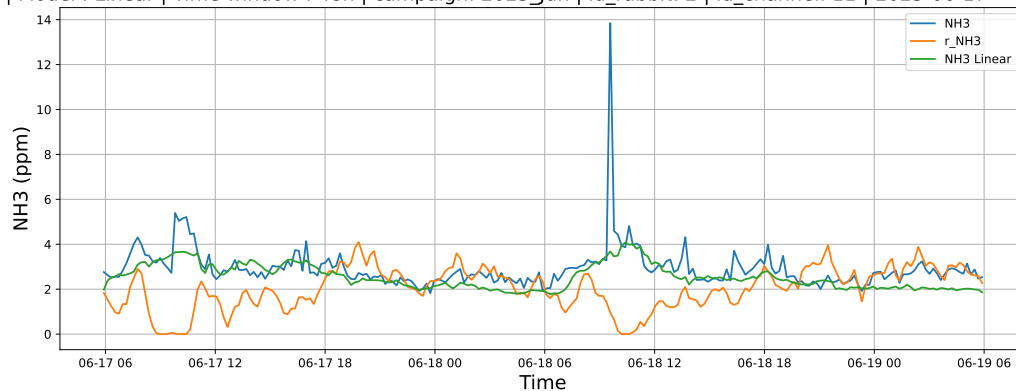


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

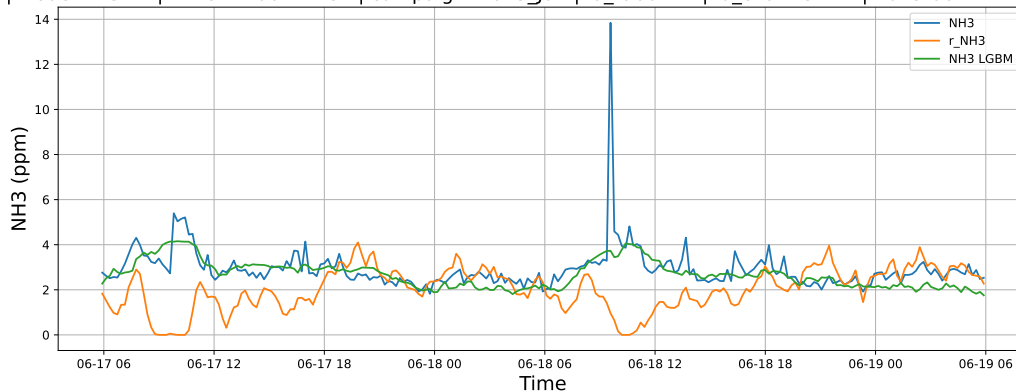


4.1.2.2 Time window : 48

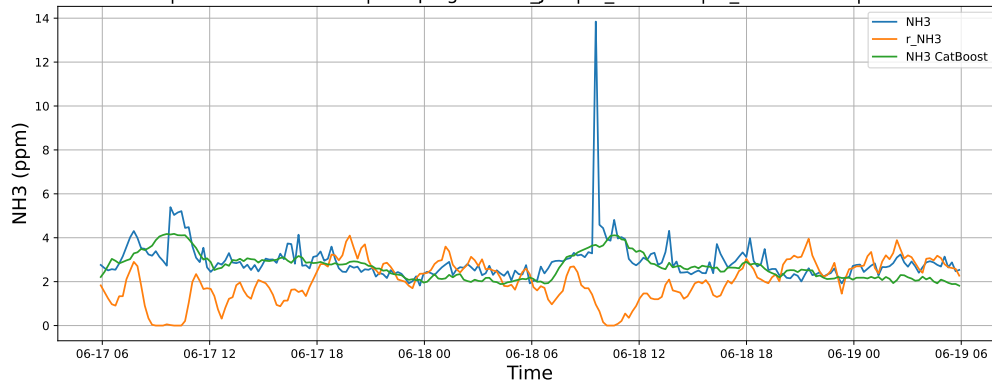
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

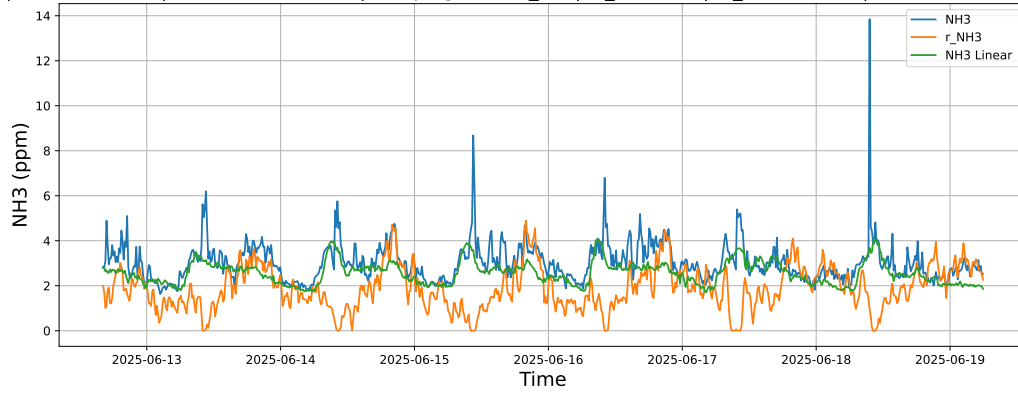


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

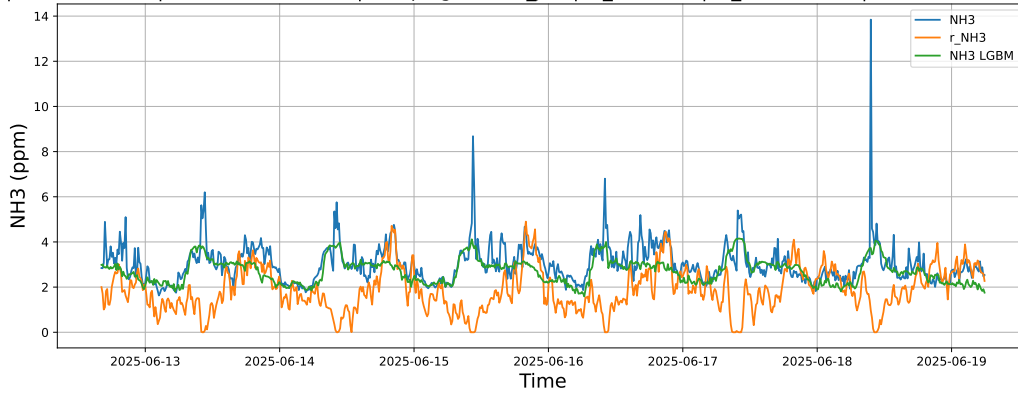


4.1.2.3 Time window : ALL

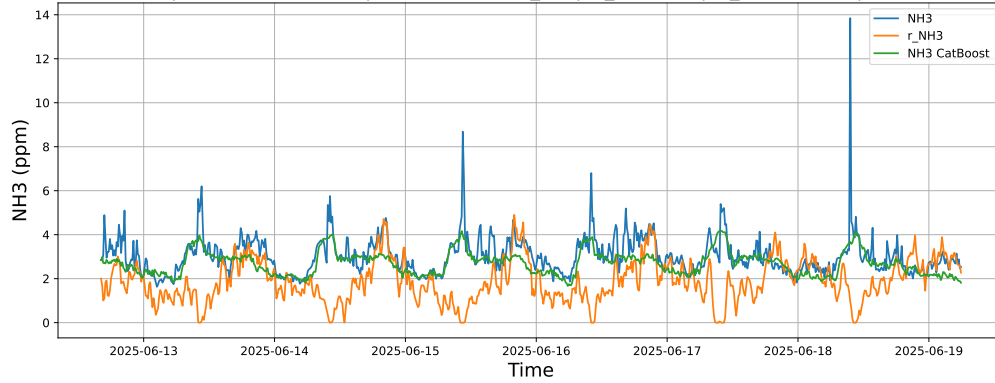
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



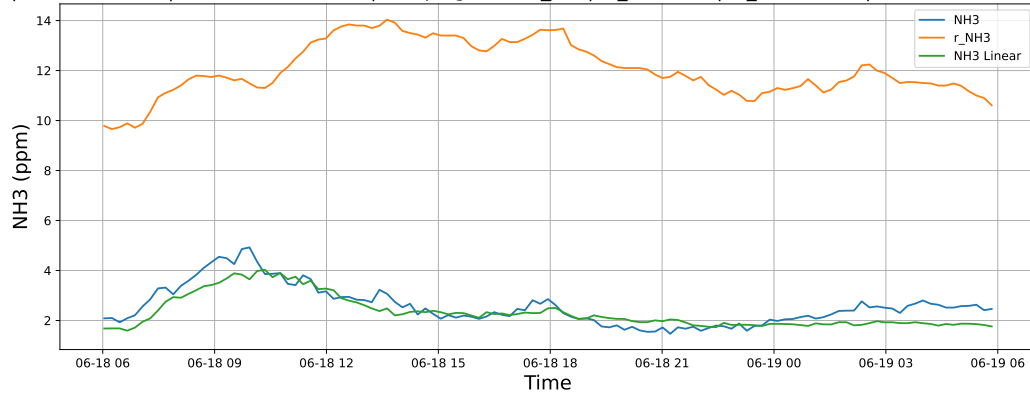
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



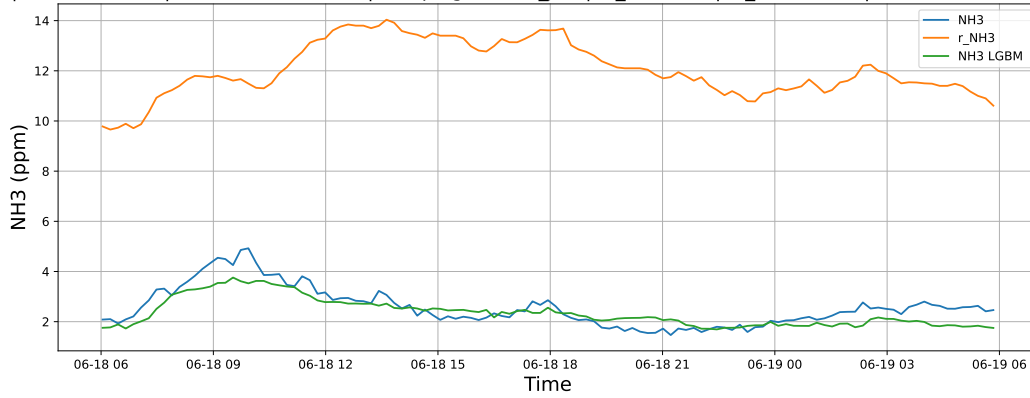
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

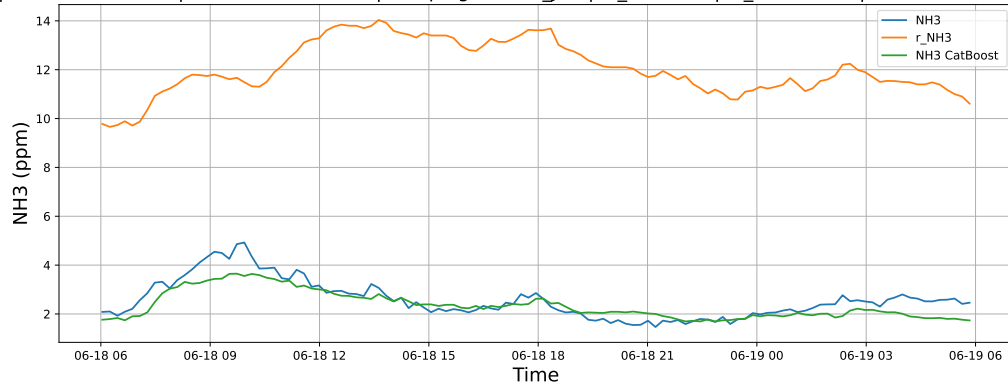
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

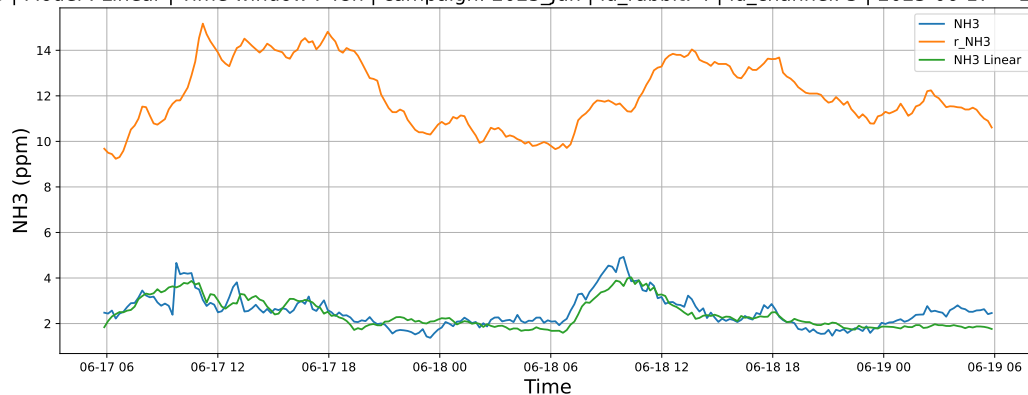


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

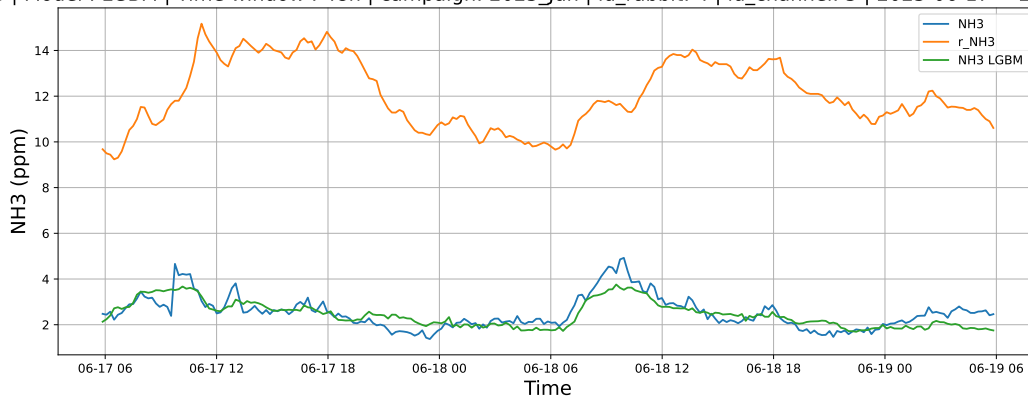


4.1.3.2 Time window : 48

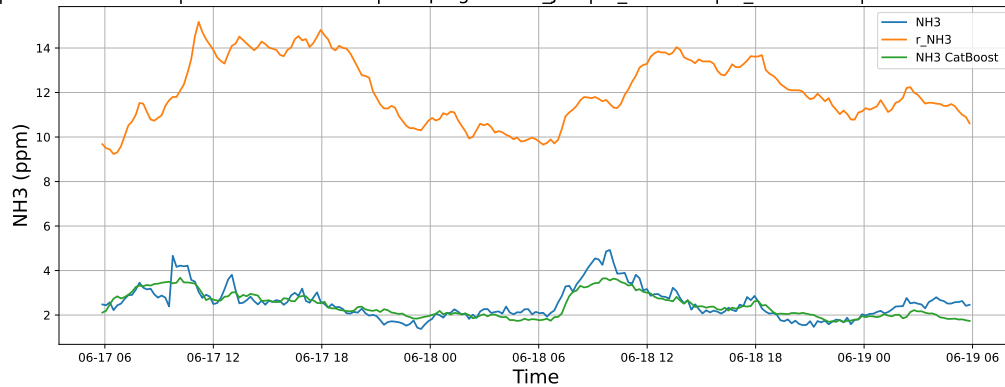
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

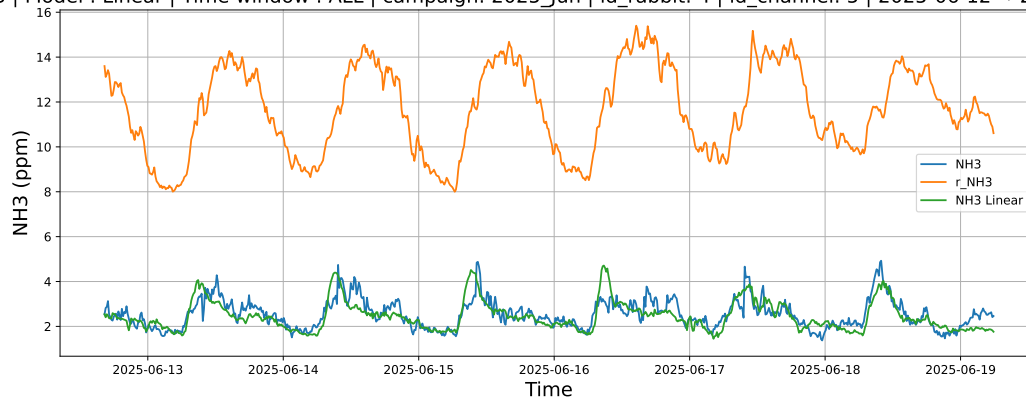


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

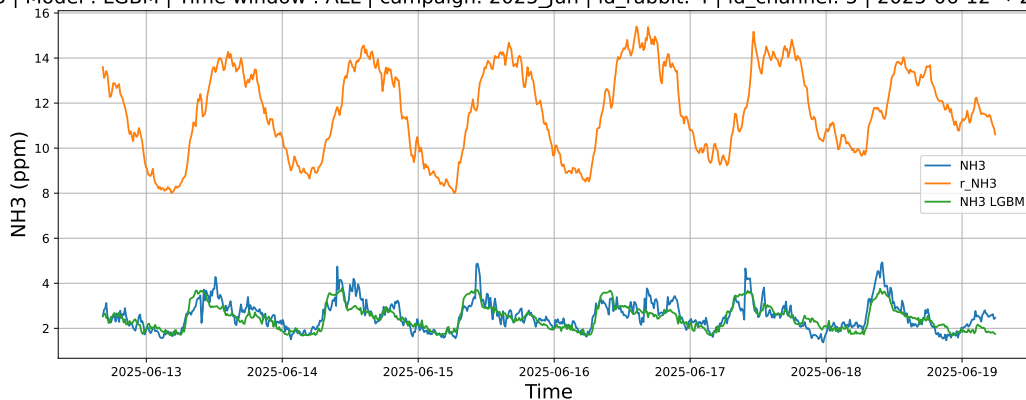


4.1.3.3 Time window : ALL

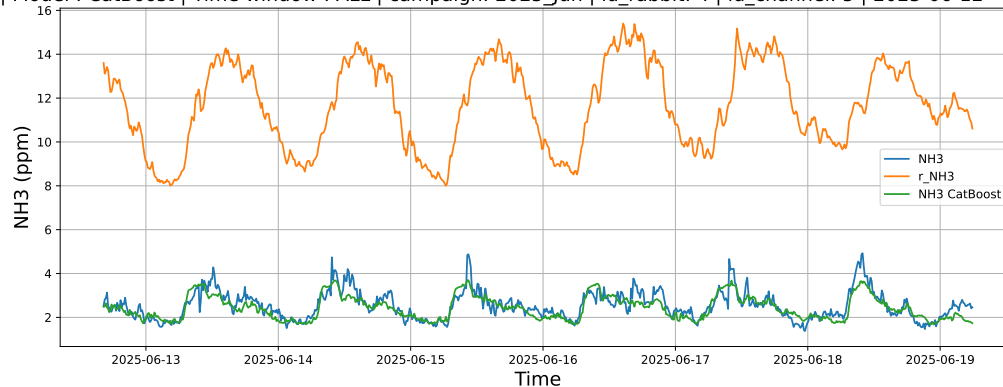
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



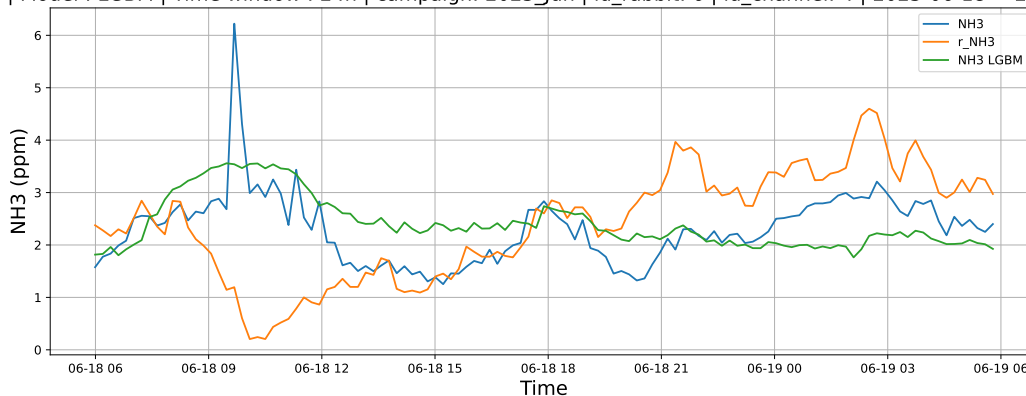
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

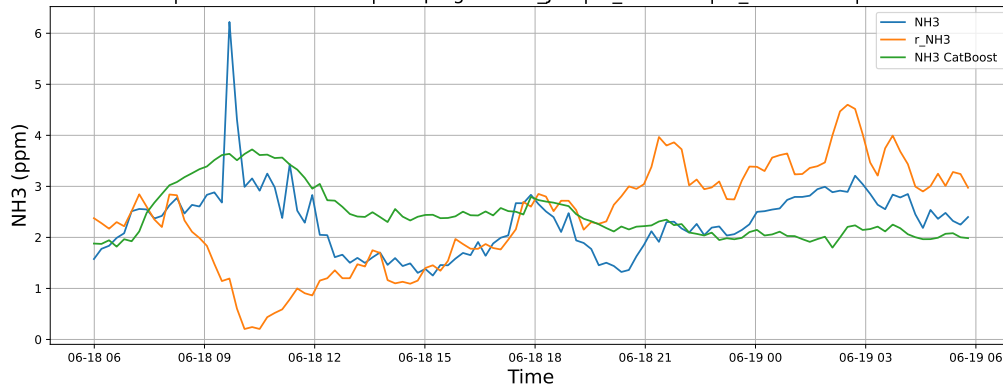
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

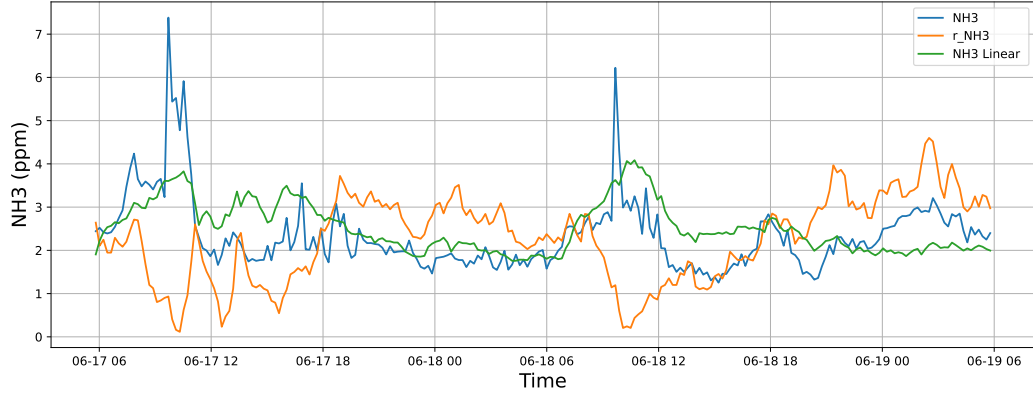


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

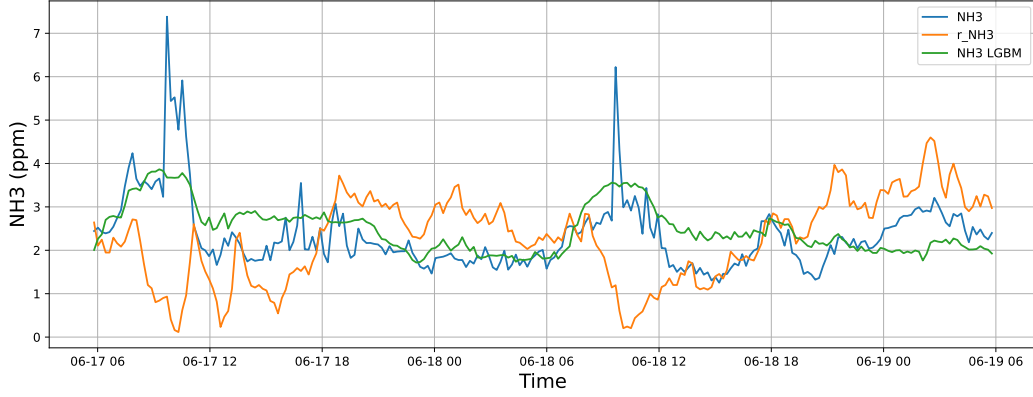


4.1.4.2 Time window : 48

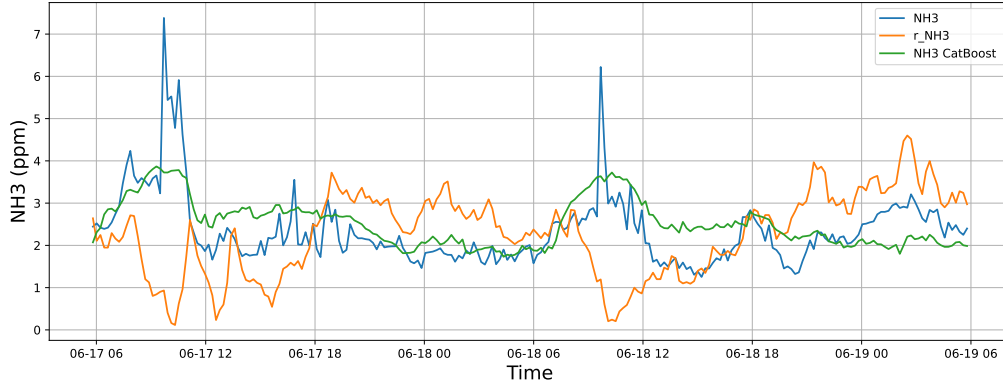
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

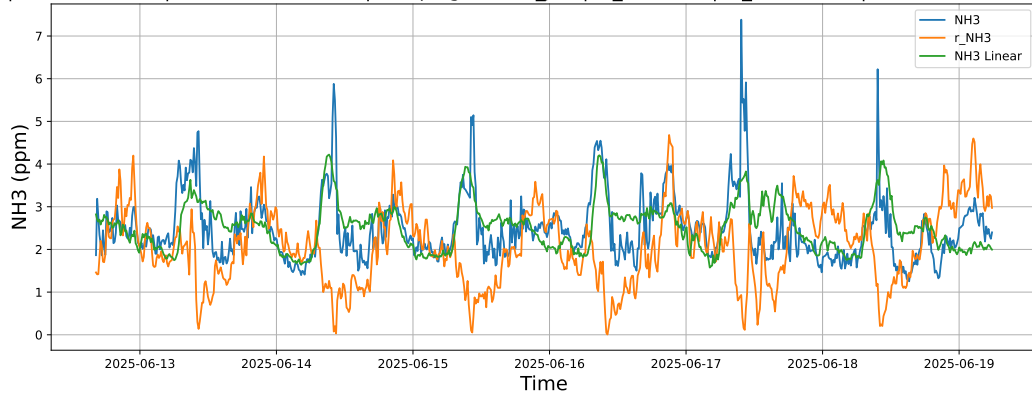


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

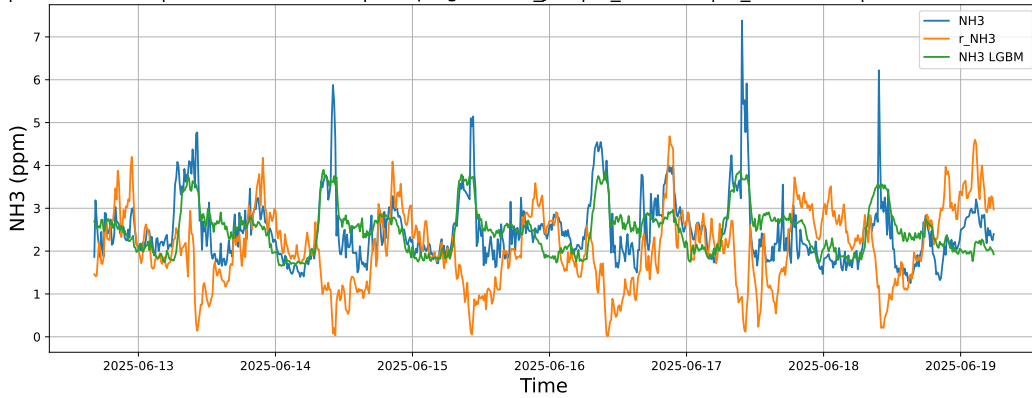


4.1.4.3 Time window : ALL

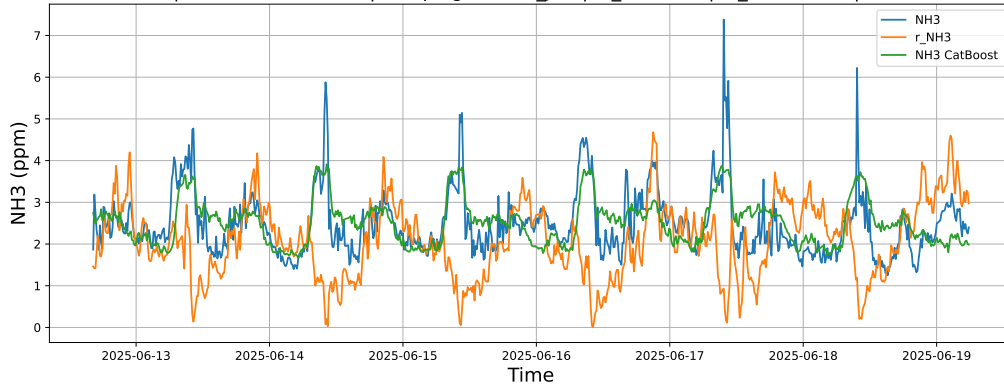
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



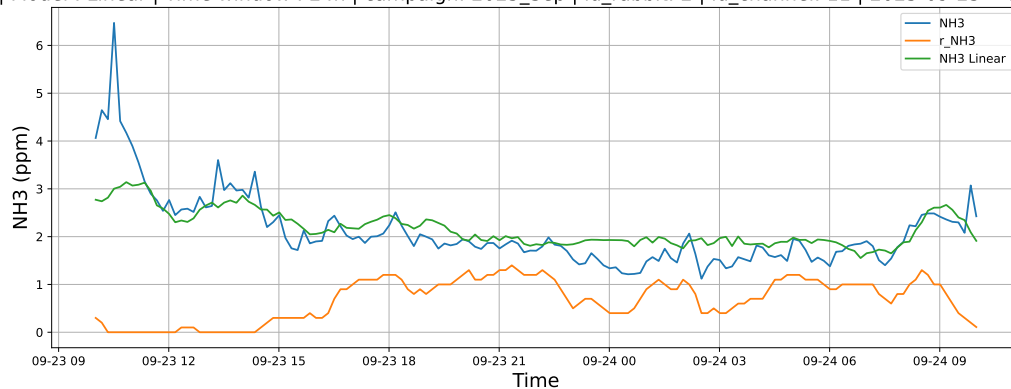
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



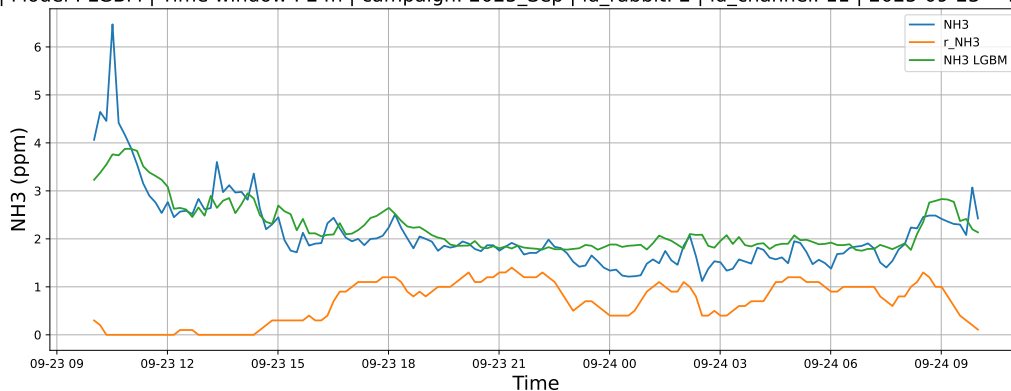
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

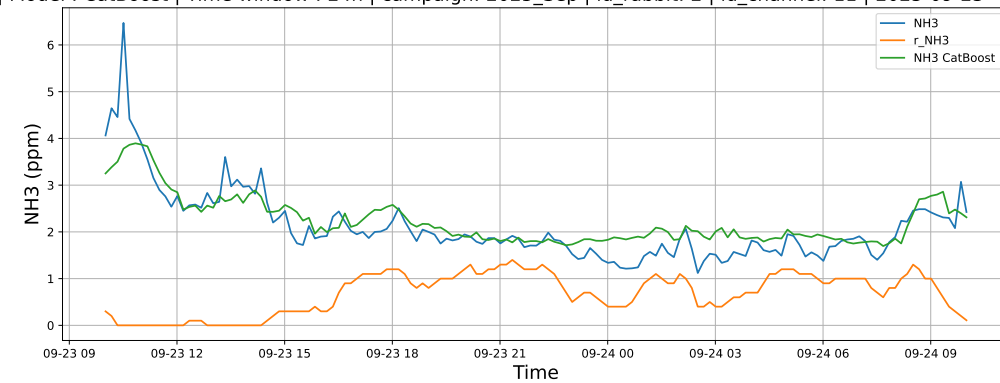
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

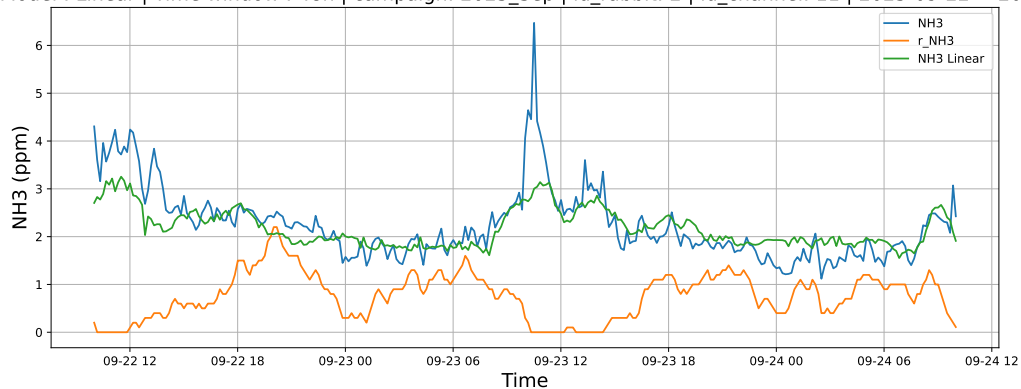


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

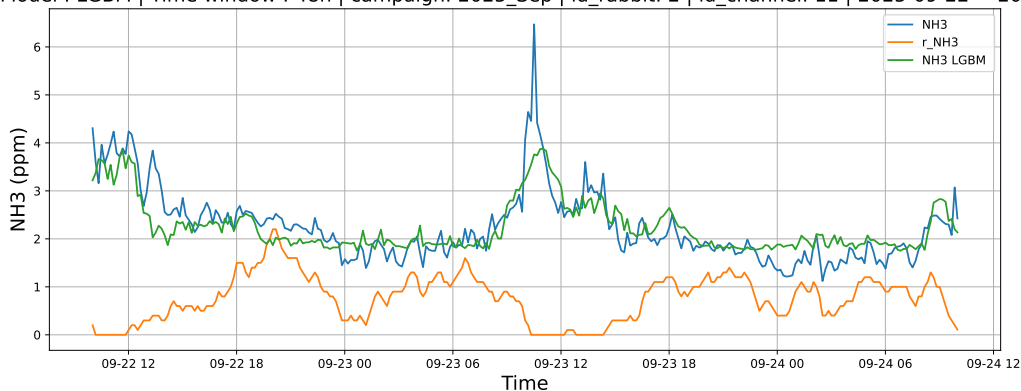


4.1.5.2 Time window : 48

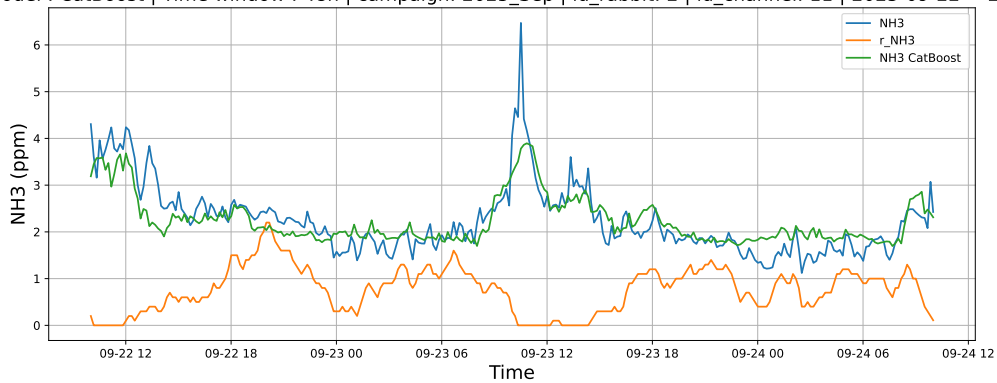
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

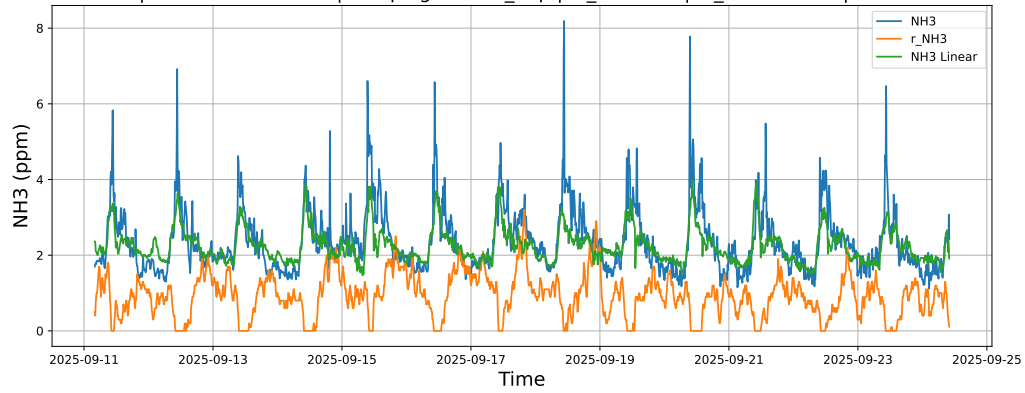


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

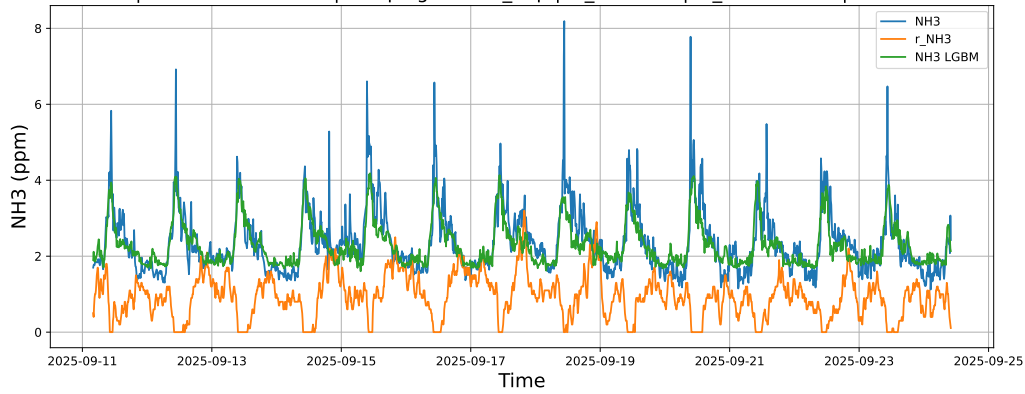


4.1.5.3 Time window : ALL

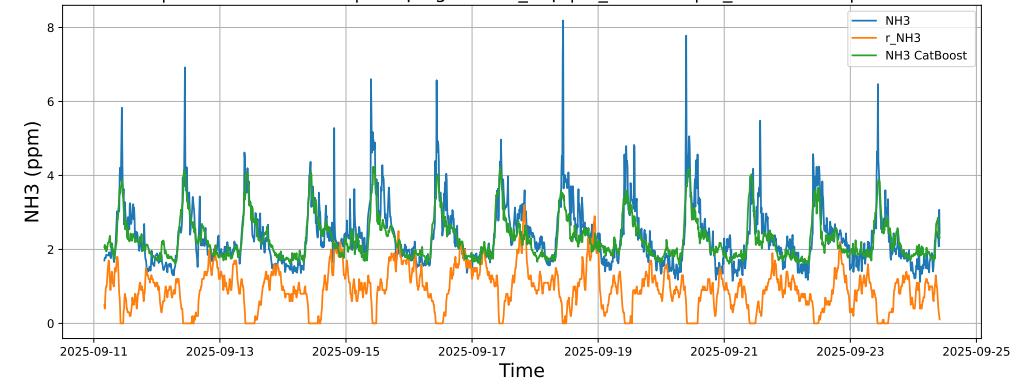
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



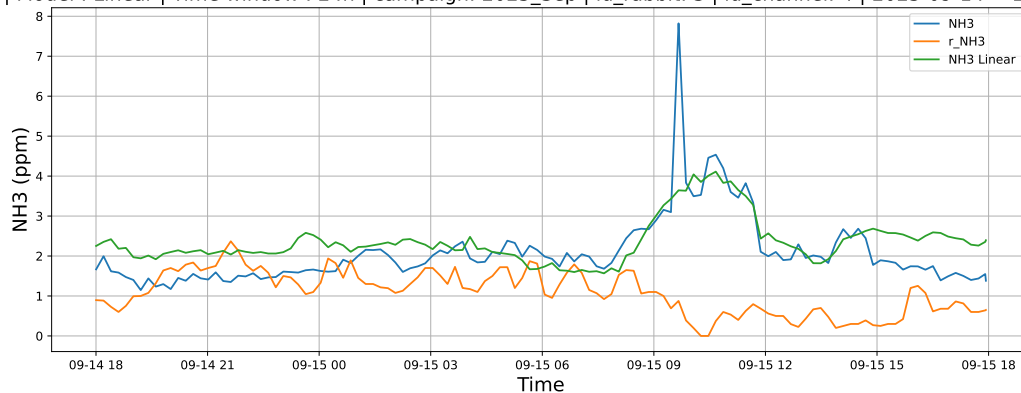
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



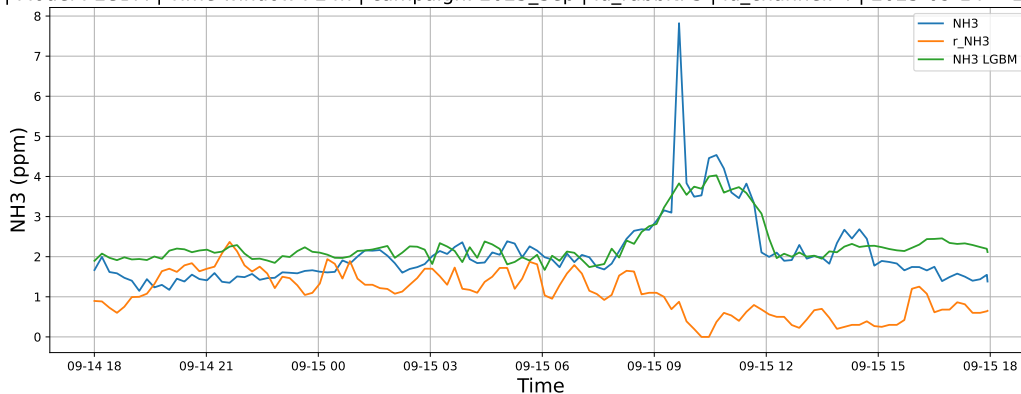
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

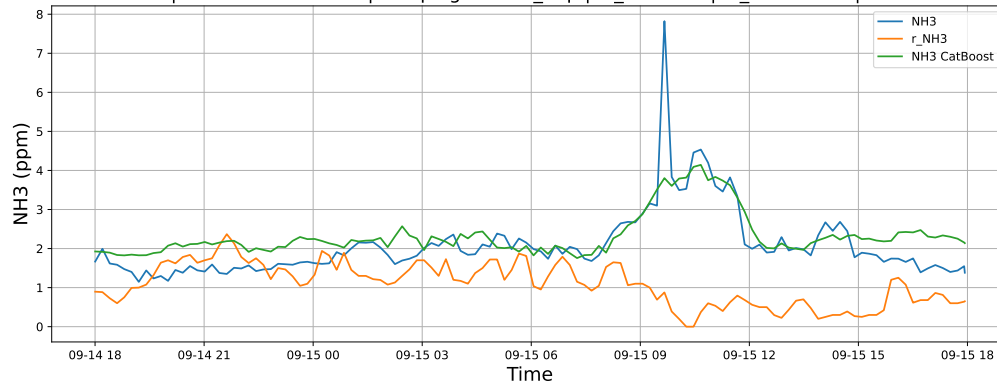
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

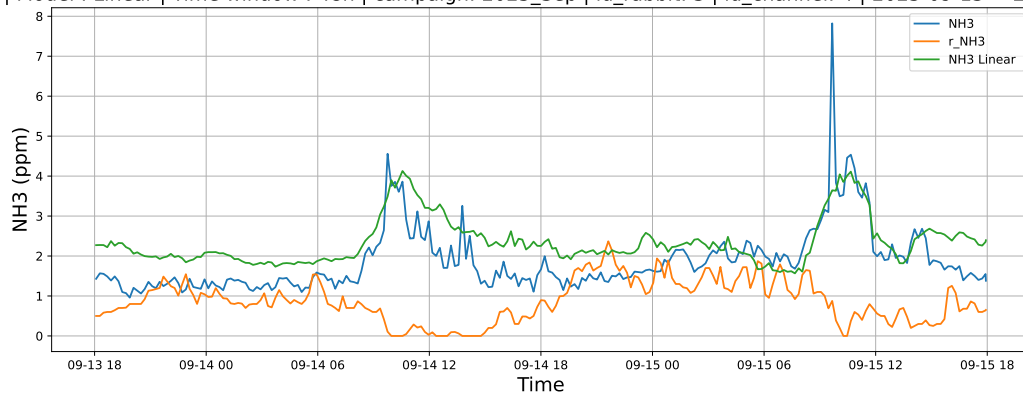


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

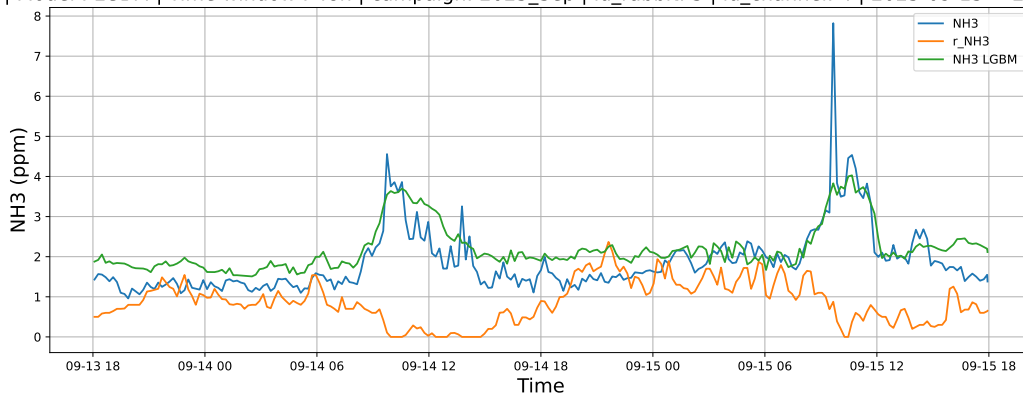


4.1.6.2 Time window : 48

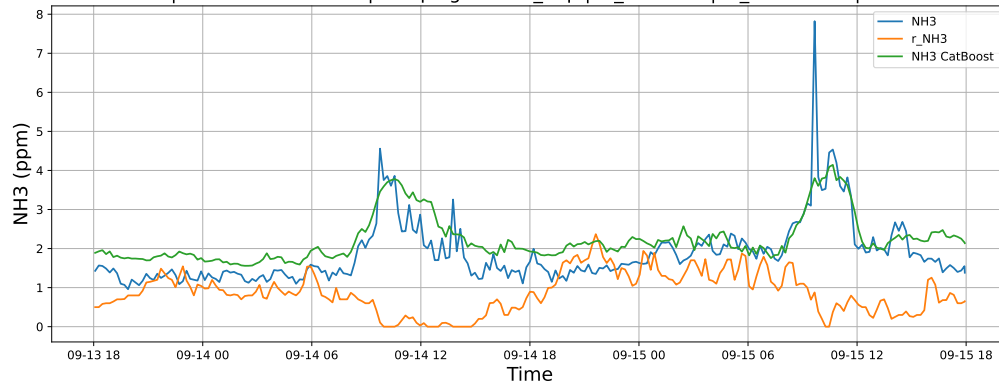
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

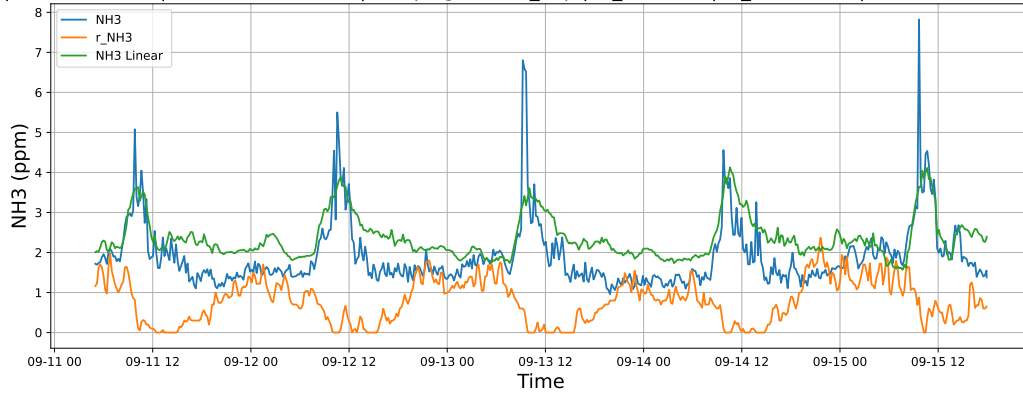


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

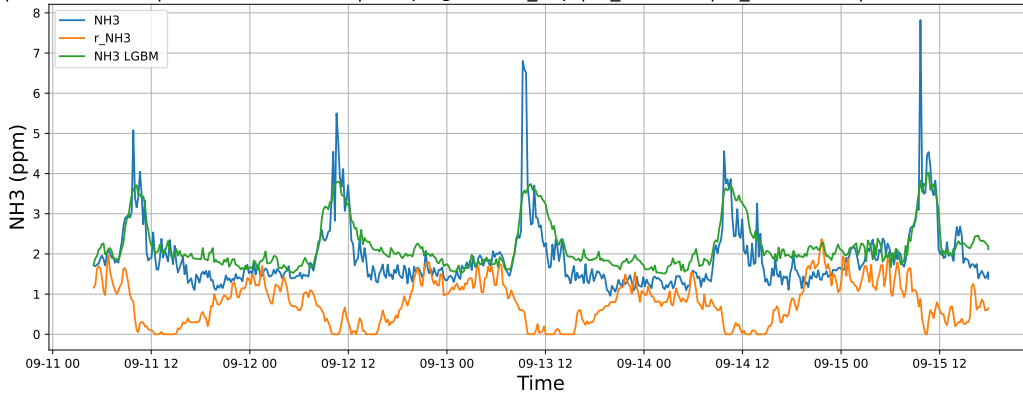


4.1.6.3 Time window : ALL

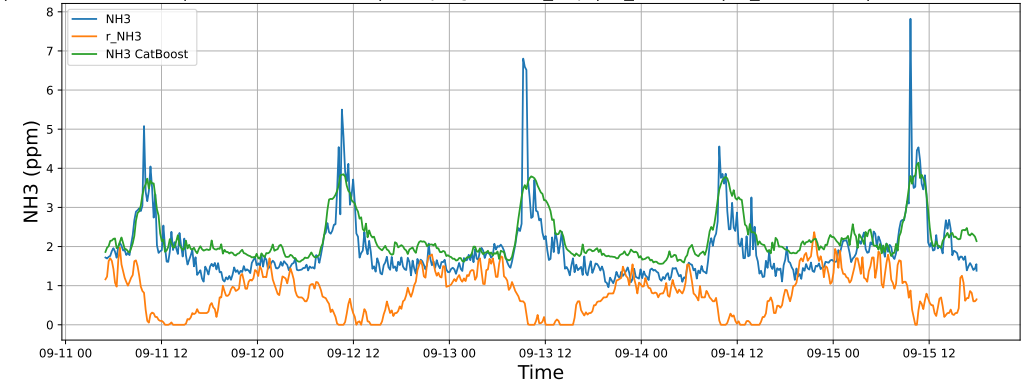
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



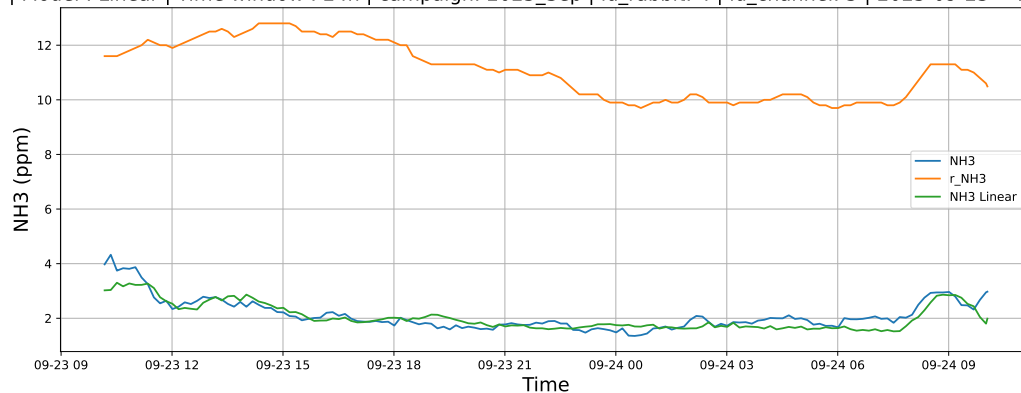
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



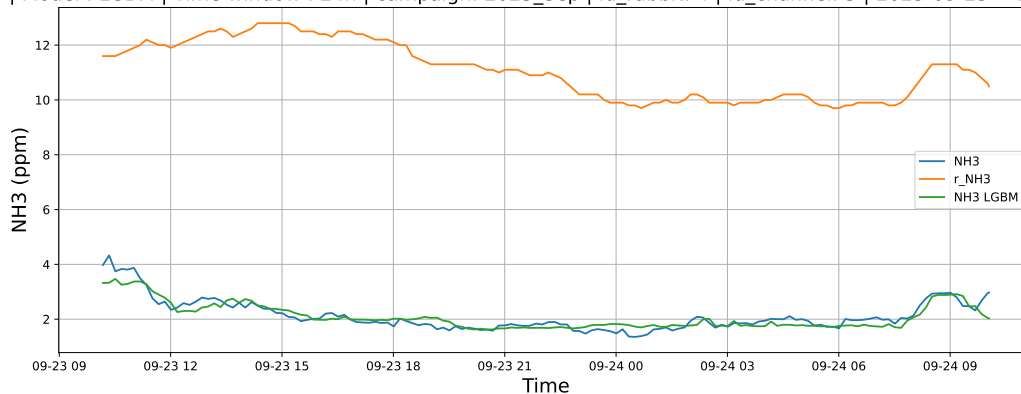
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

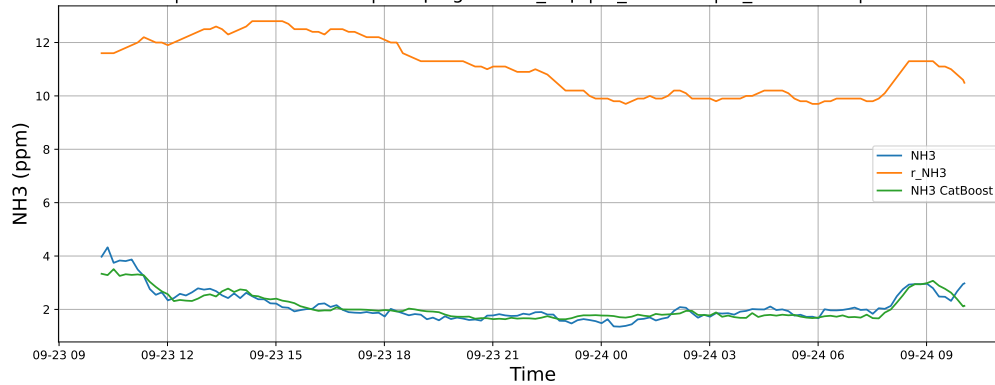
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

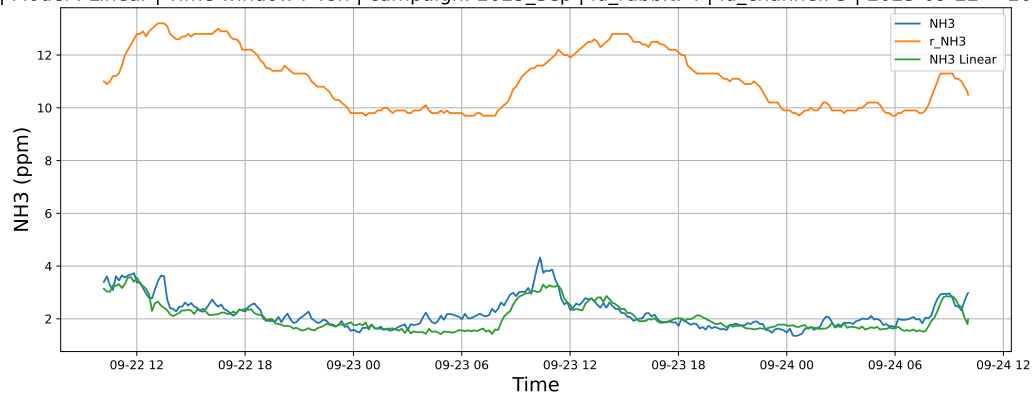


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

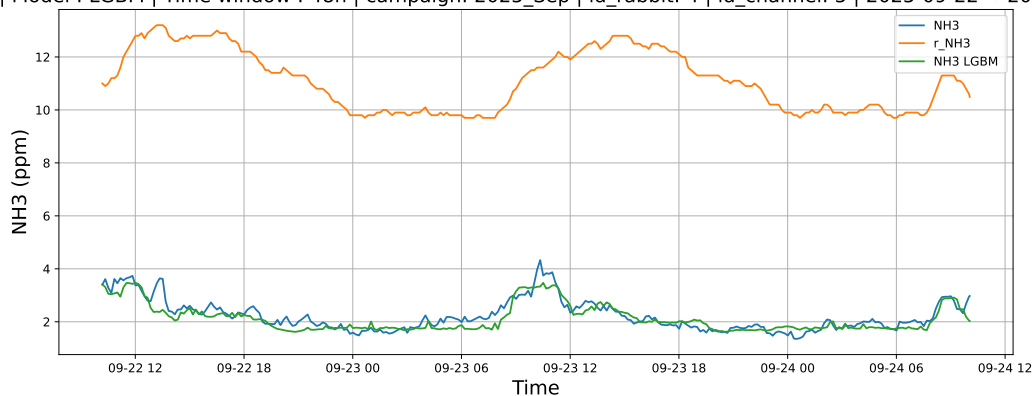


4.1.7.2 Time window : 48

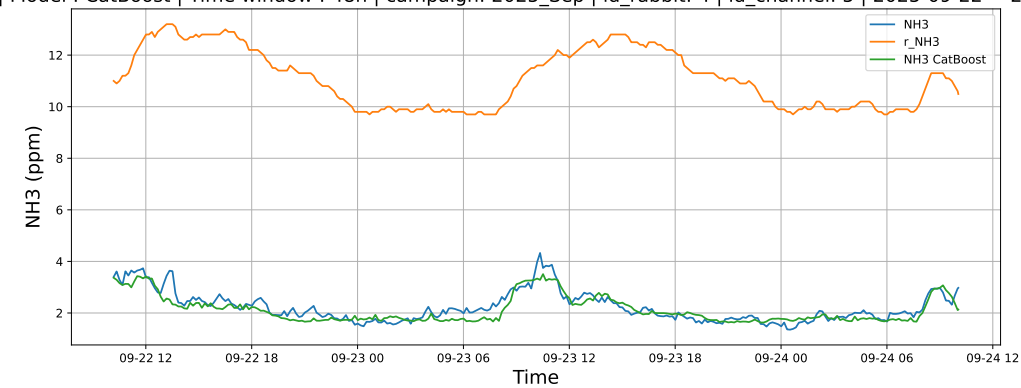
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

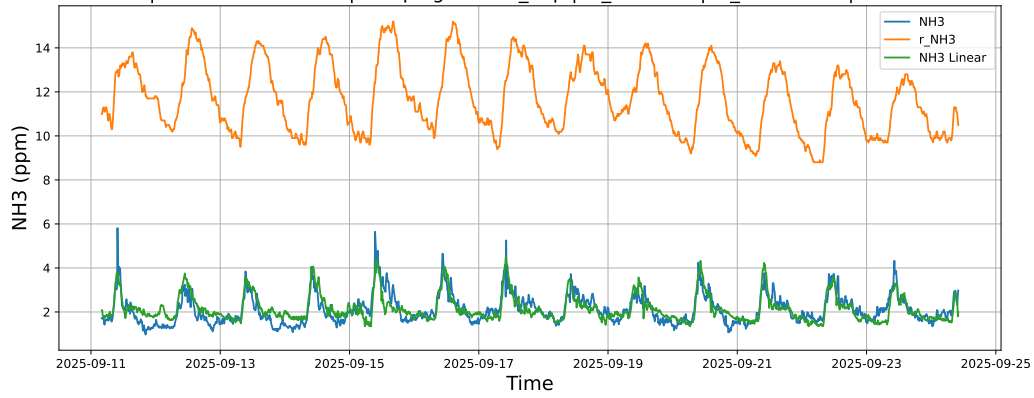


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

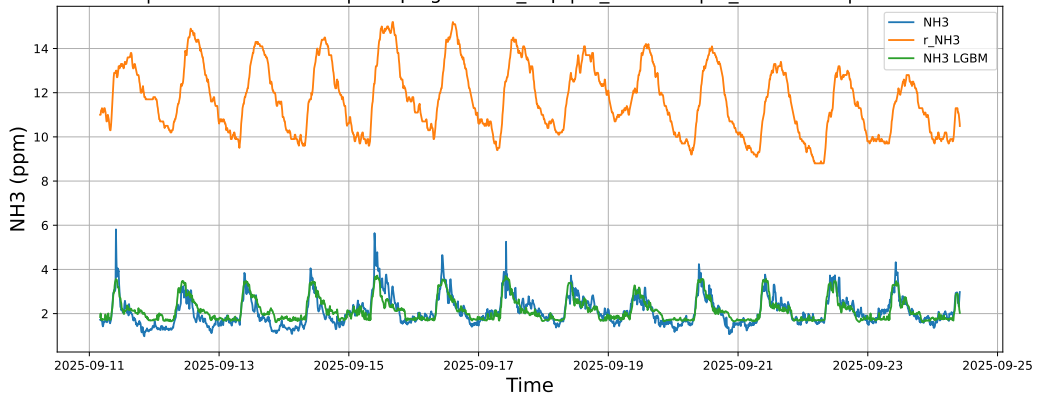


4.1.7.3 Time window : ALL

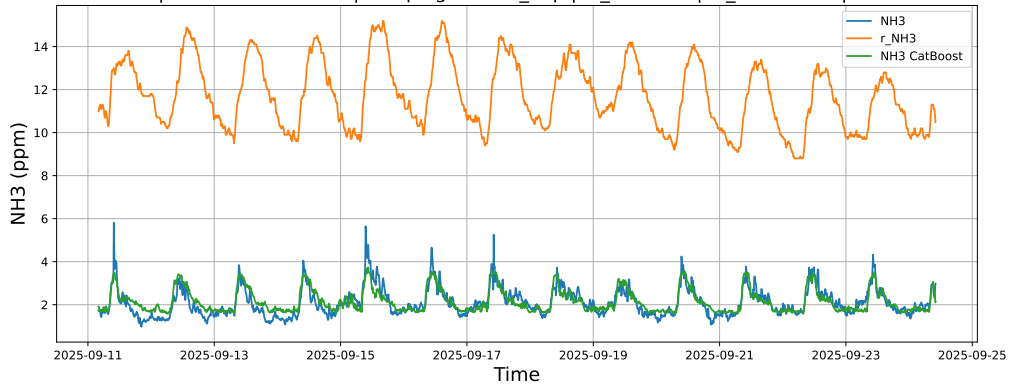
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



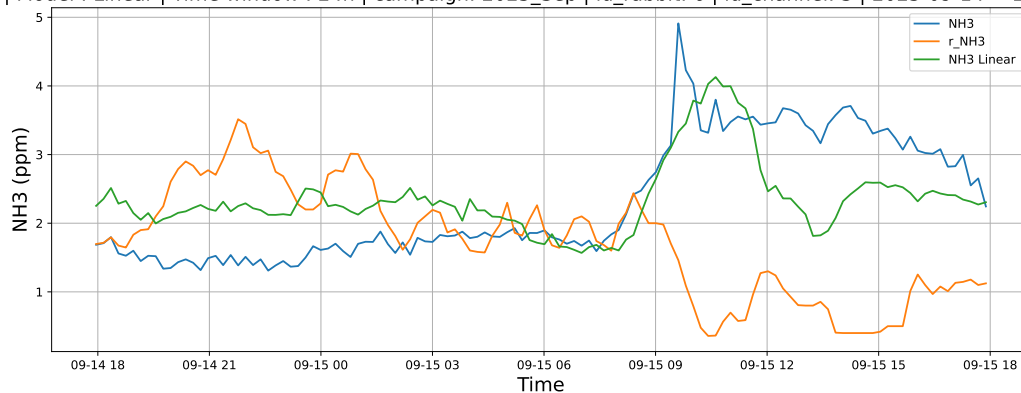
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



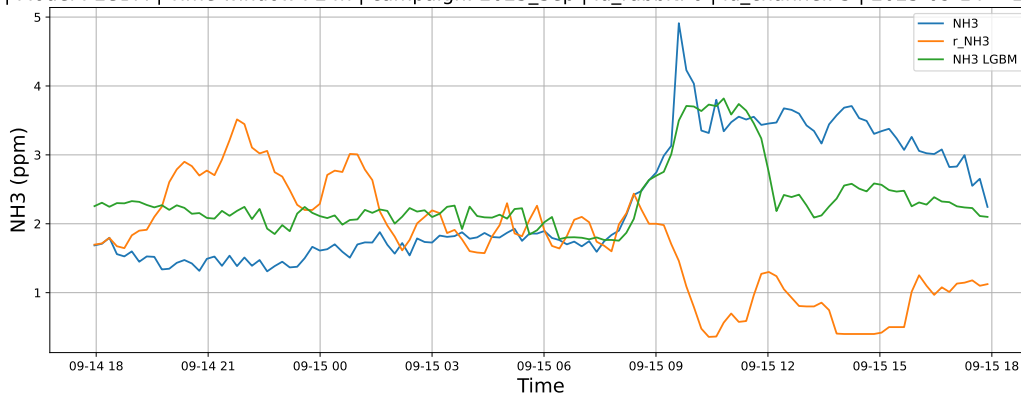
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

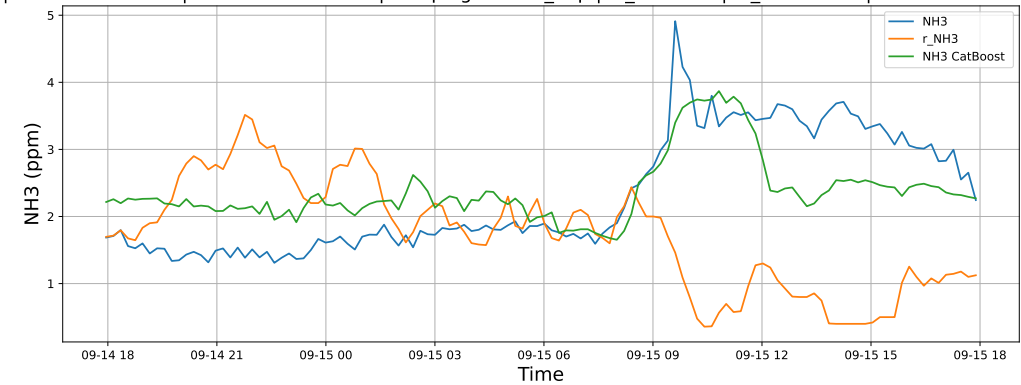
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

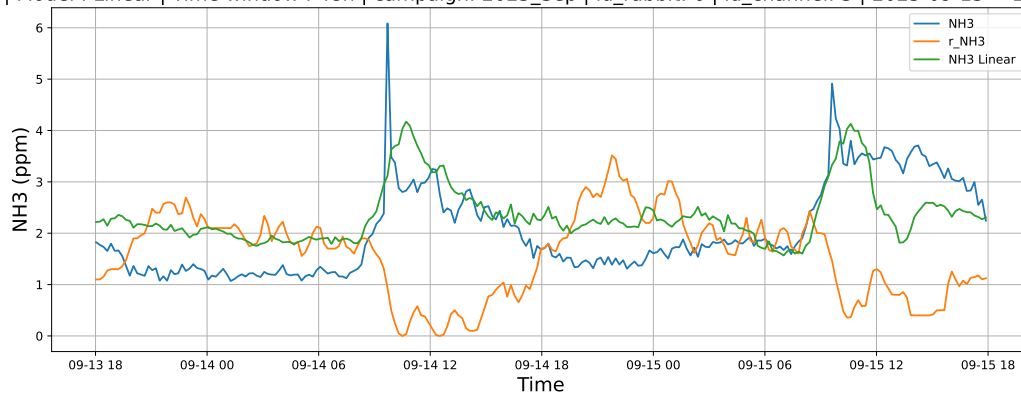


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

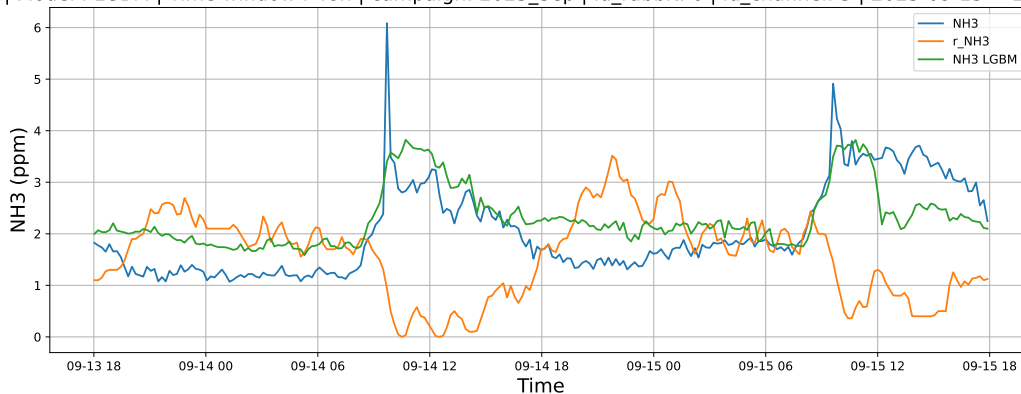


4.1.8.2 Time window : 48

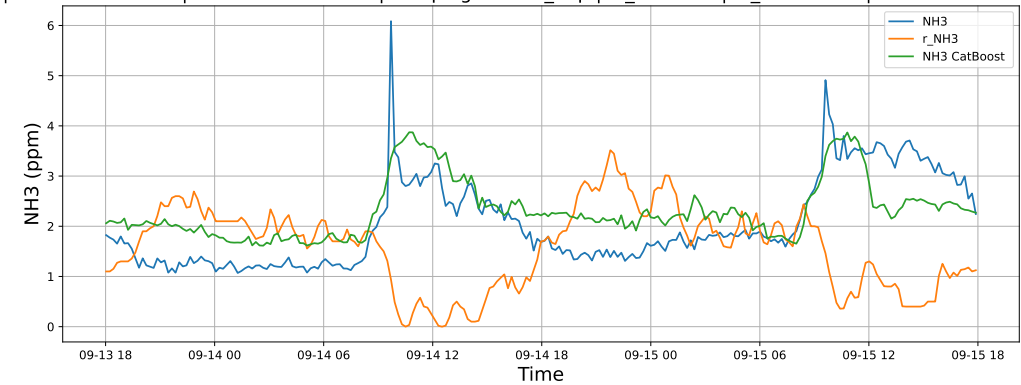
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

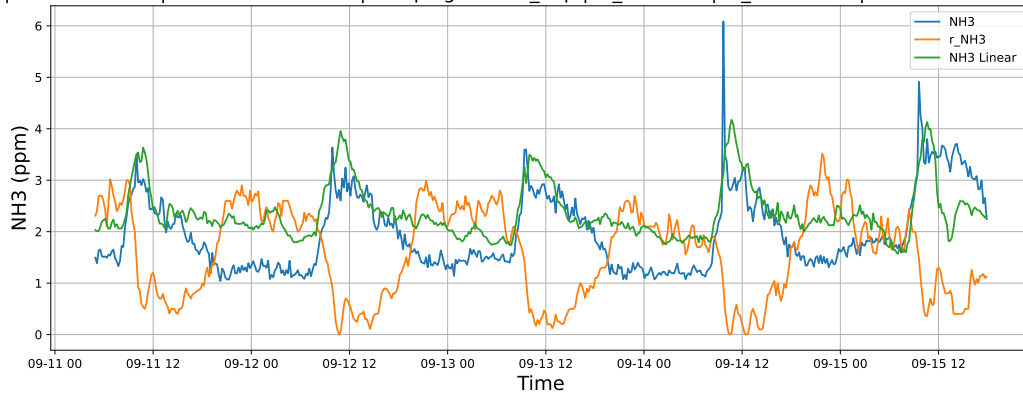


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

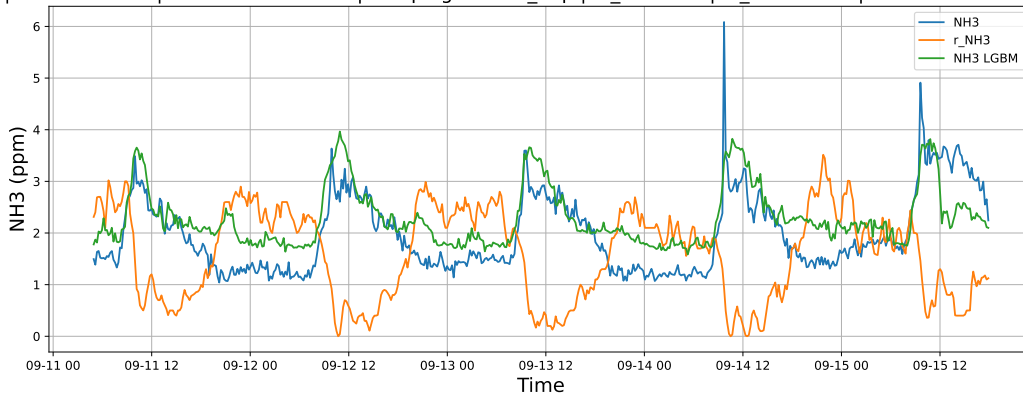


4.1.8.3 Time window : ALL

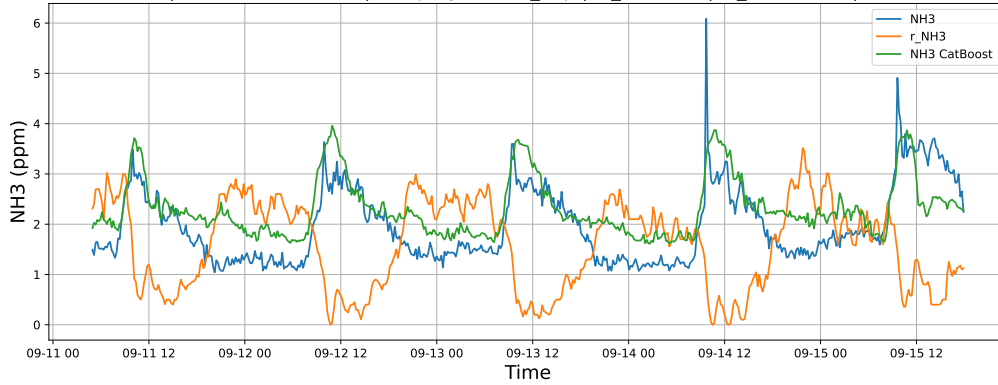
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

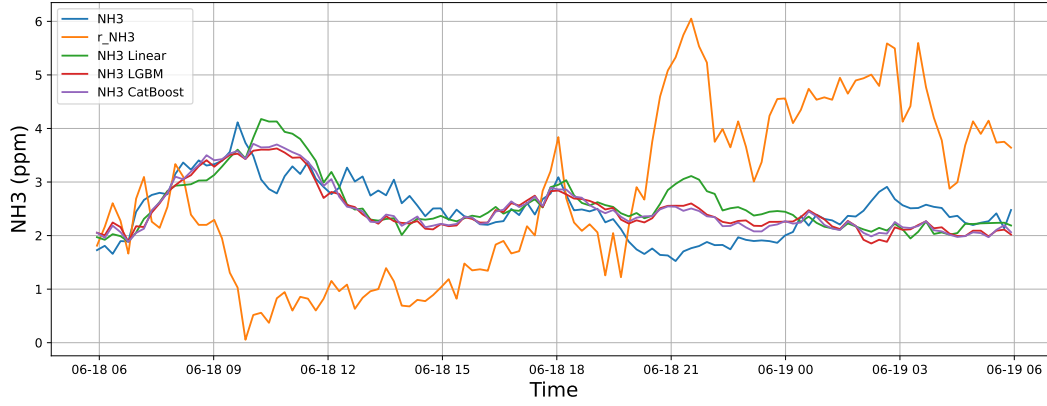


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

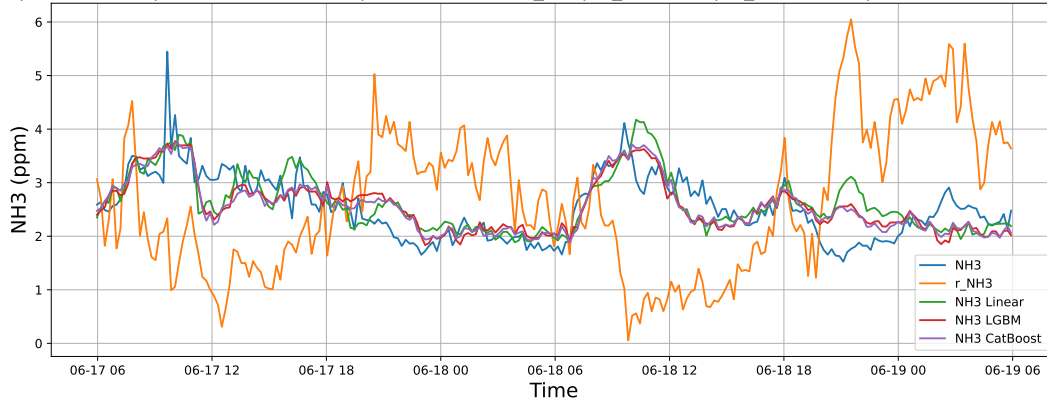
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



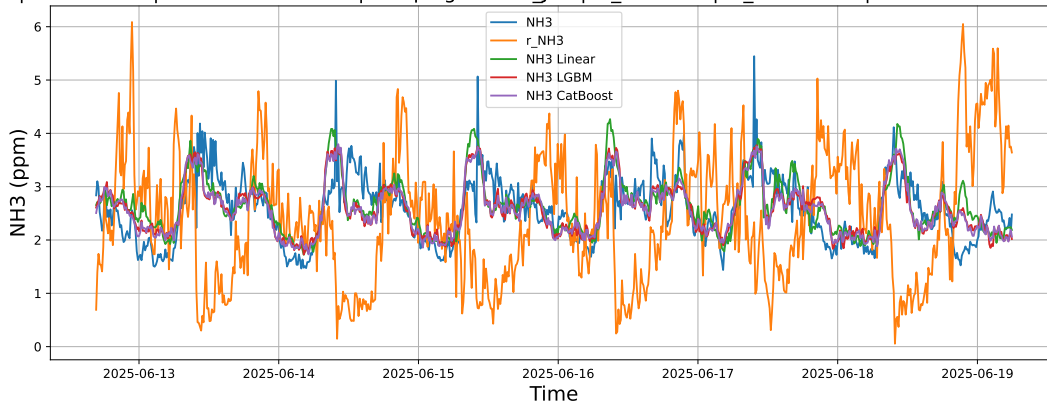
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

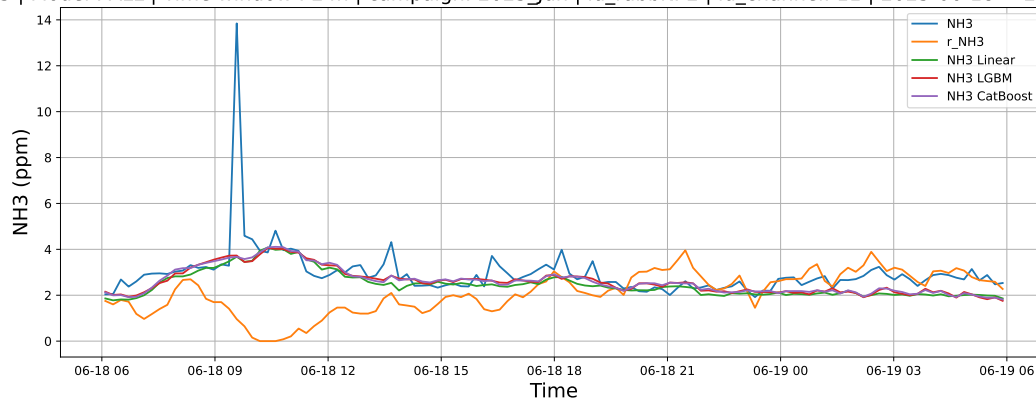
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

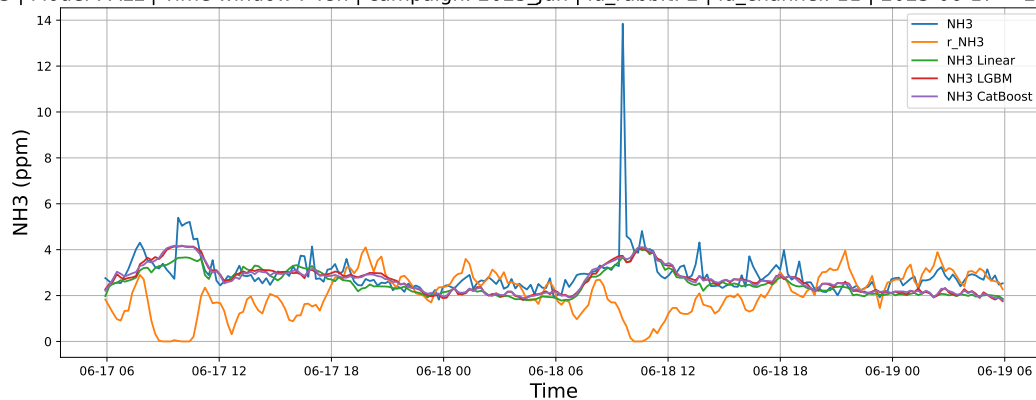
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



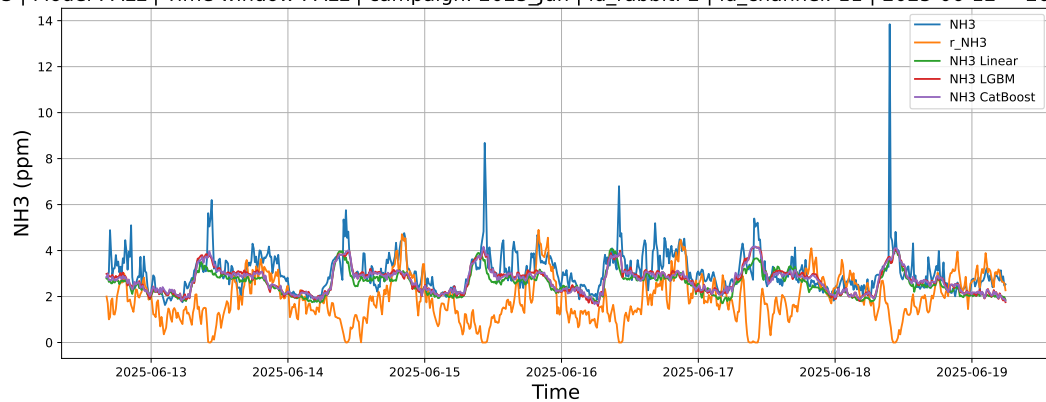
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

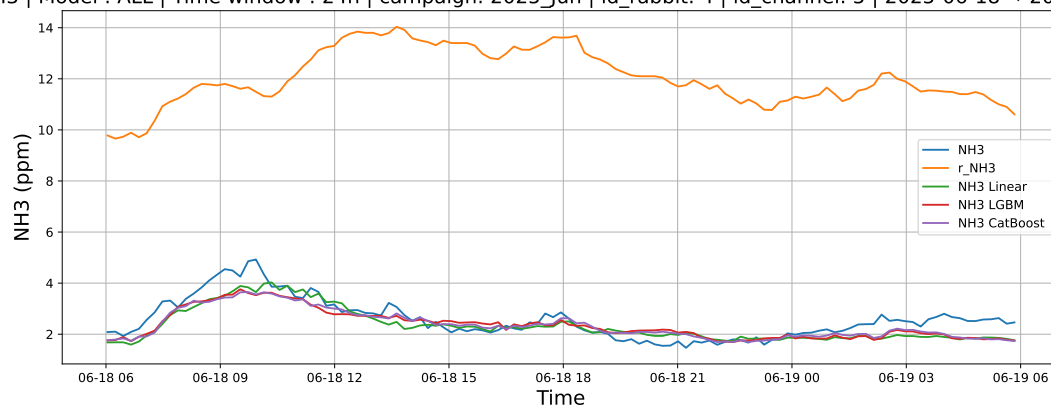
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

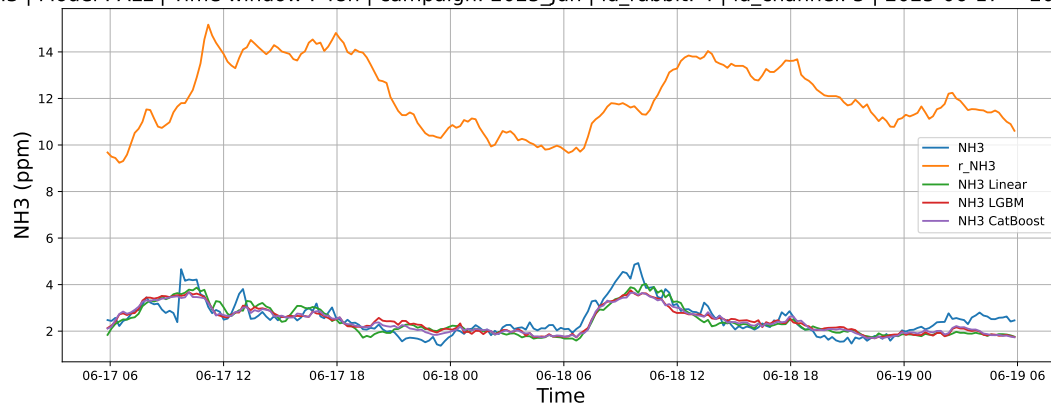
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



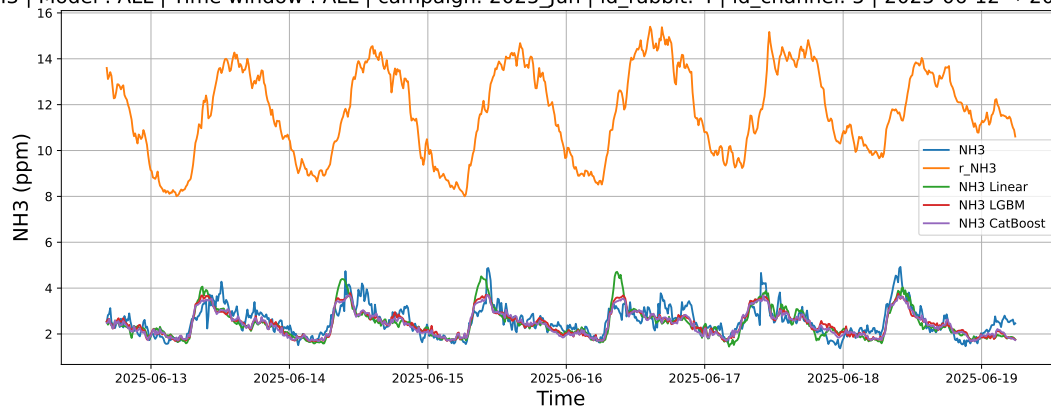
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

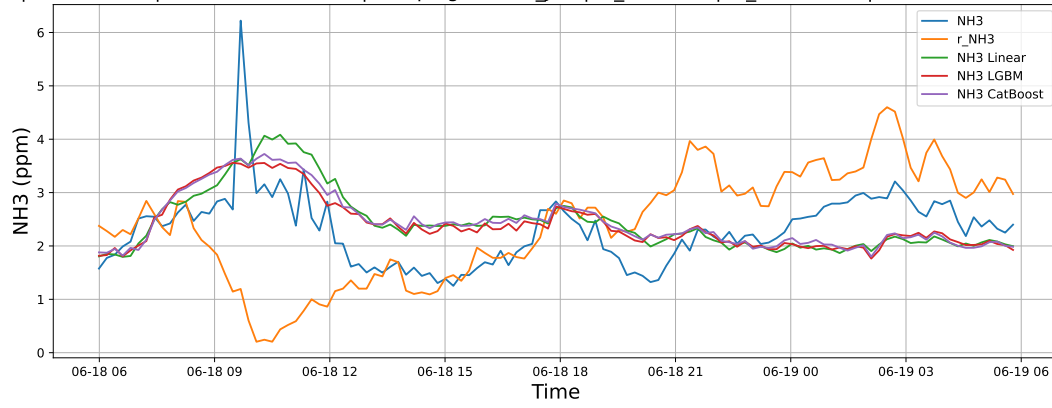
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

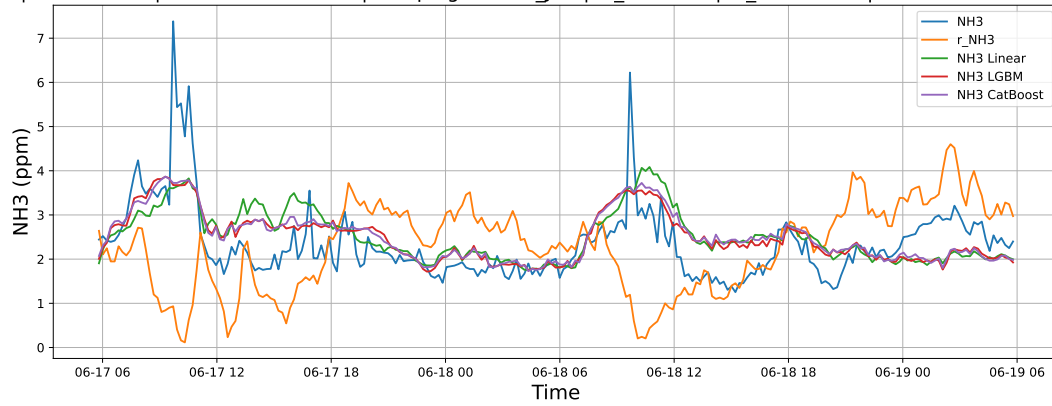
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



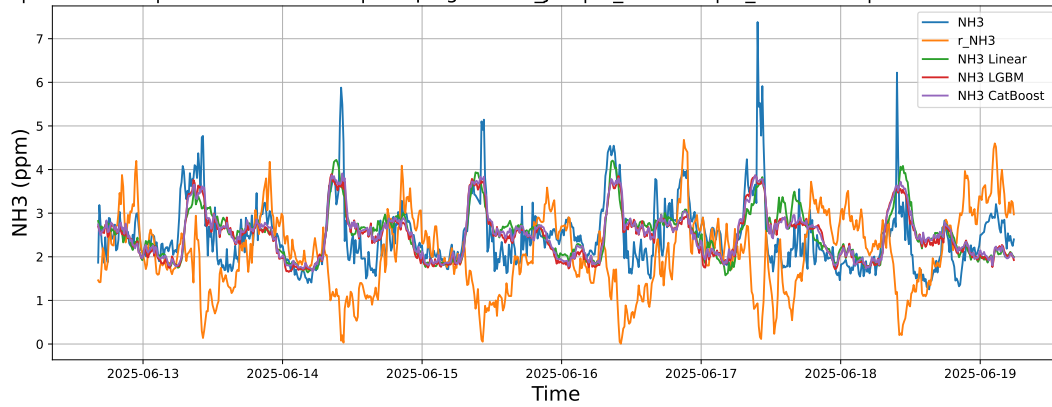
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

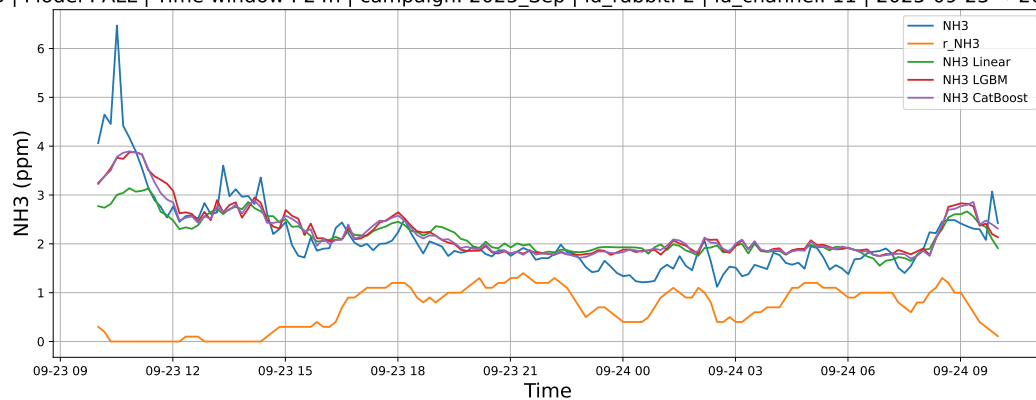
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

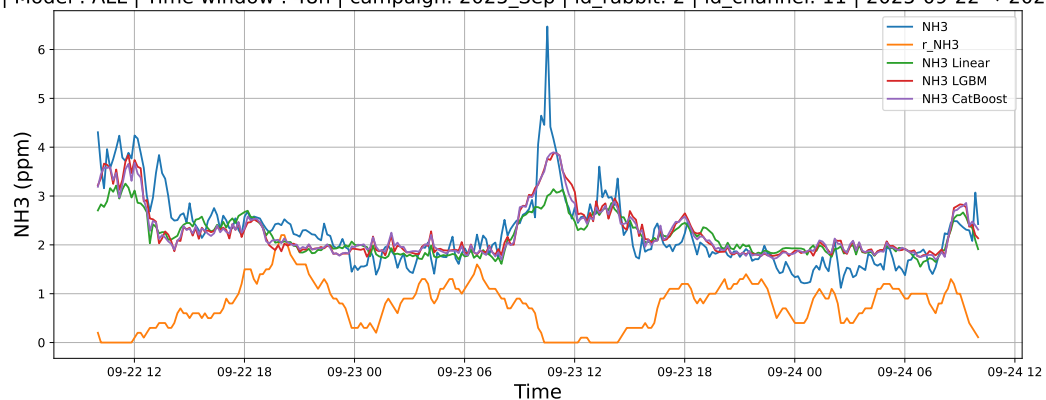
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



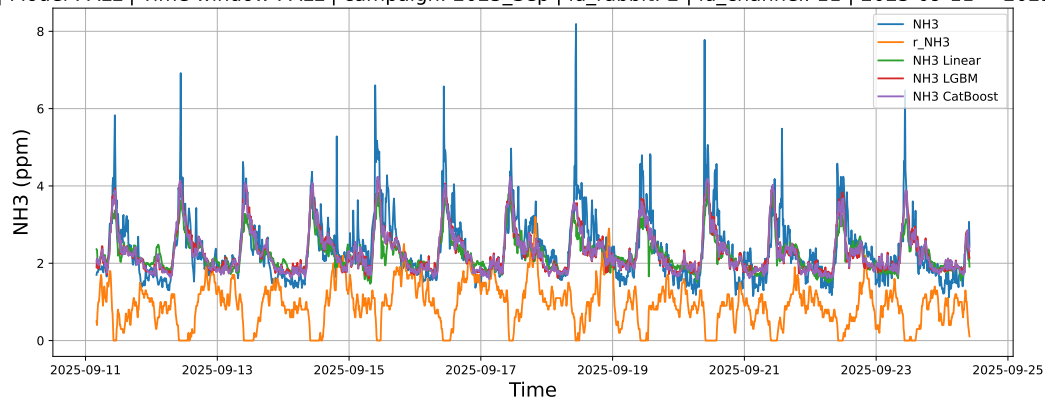
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

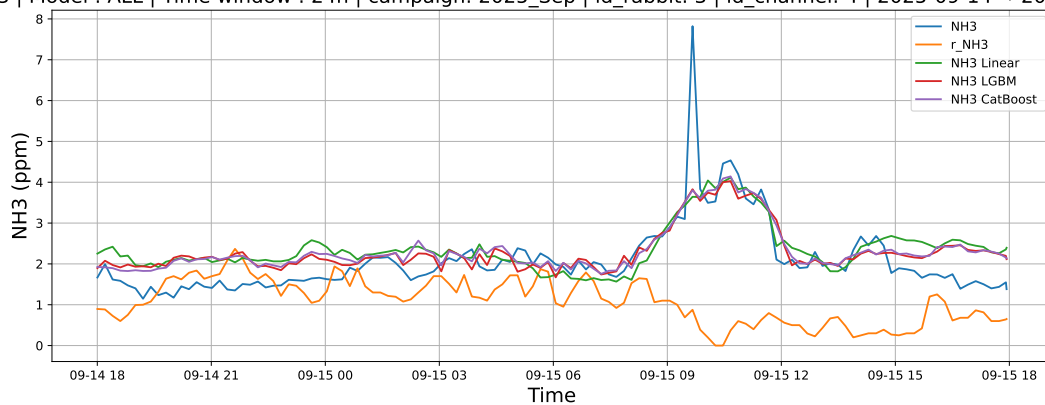
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

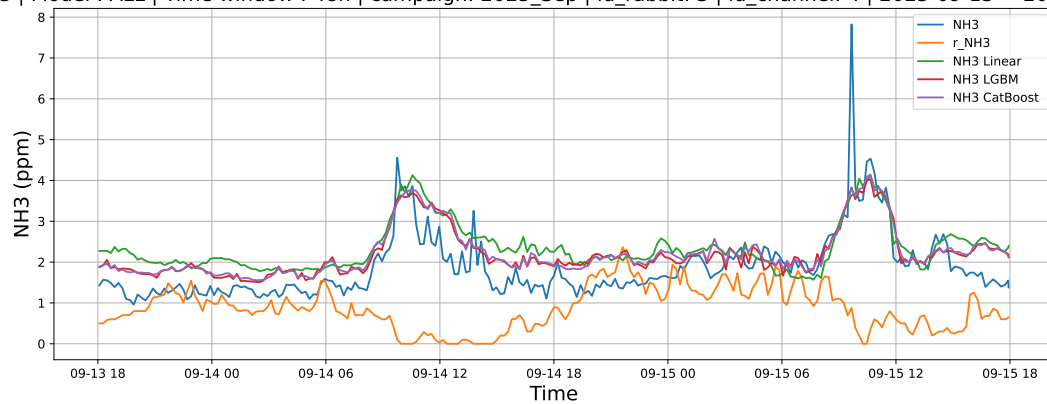
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



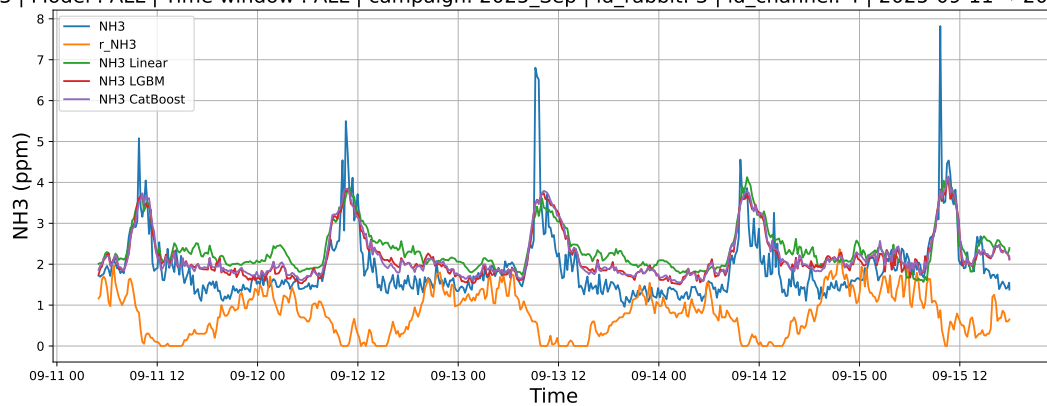
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

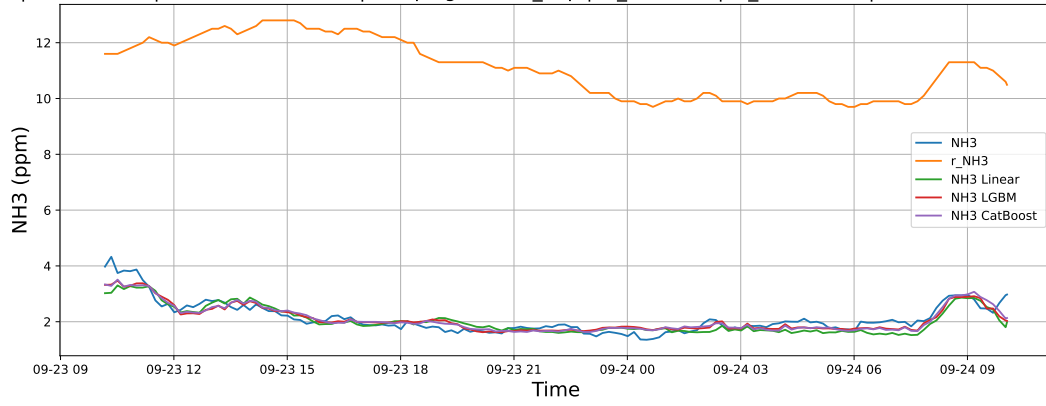
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

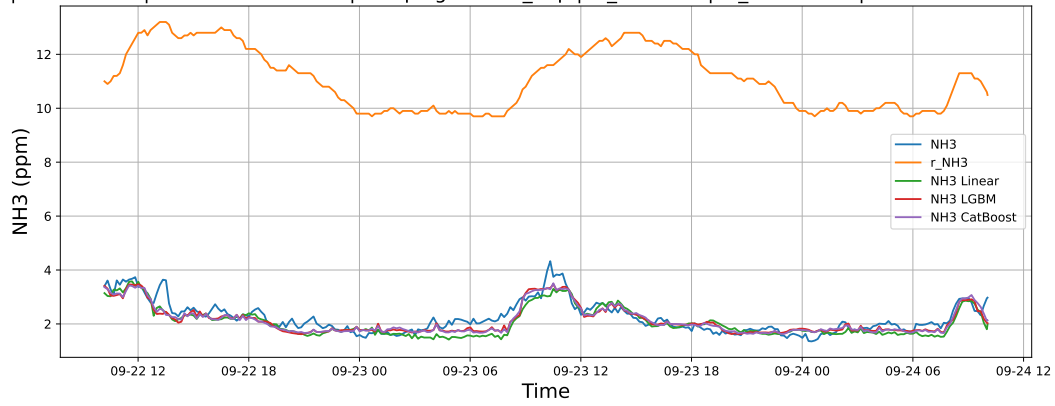
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



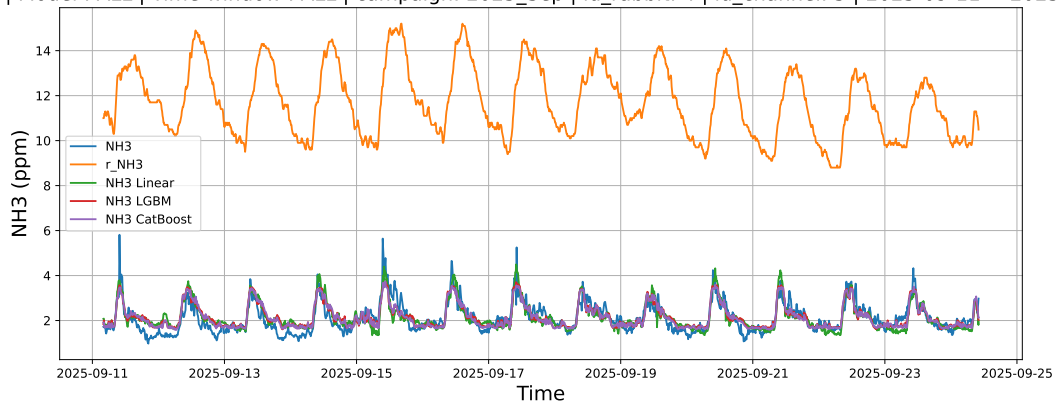
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

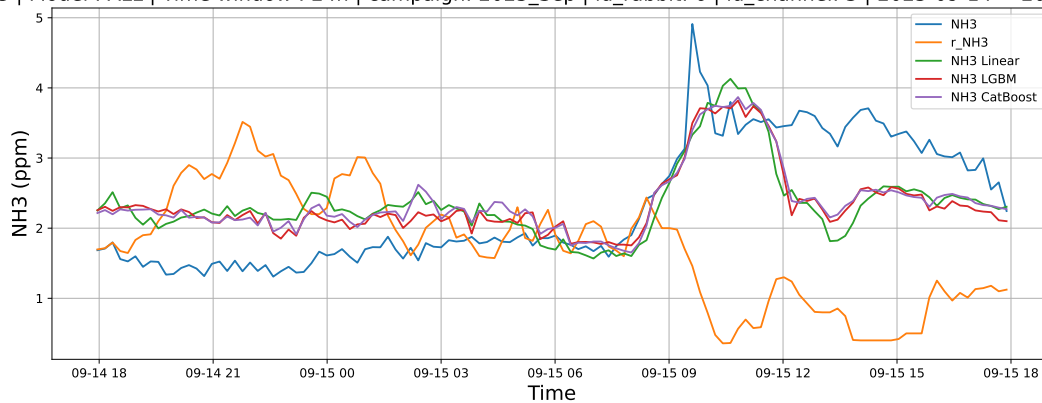
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

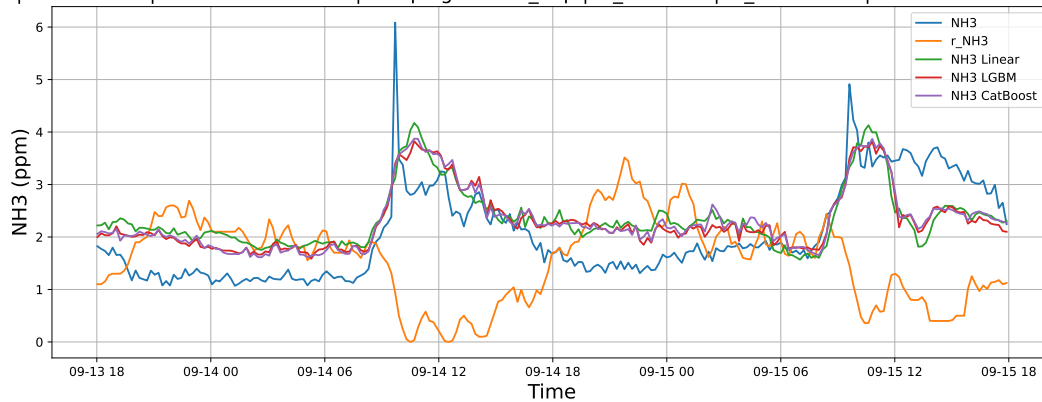
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

